

**GRID CONNECTION CODE FOR BATTERY ENERGY
STORAGE FACILITIES (*BESF*) CONNECTED TO THE
ELECTRICITY *TRANSMISSION SYSTEM (TS)* OR
THE *DISTRIBUTION SYSTEM (DS)* IN SOUTH
AFRICA**

Version 5.3

**This document is approved by the National Energy Regulator of
South Africa (NERSA)**

Administered by:

The SA Grid Code Secretariat

Contact: Mr. T. Mchunu

System Operator, Eskom Transmission Division

P.O Box 103, Germiston 1400

Tel: +27 (0)11 871 3076

Mobile: +27 (0)82 817 4542

Email: mchunut@eskom.co.za

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1. Grid Connection Code Basis

1.1. Legislation

- (1) The legal basis for this Battery Energy Storage Facilities grid connection code is specified in terms of the Electricity Regulation Act (Act 4 of 2006), as amended.
- (2) This Grid Connection Code sets the requirements for BESF connected to the Transmission System (TS) or Distribution System (DS)
- (3) The BESF Code shall be read in conjunction with all other applicable codes, rules and regulations approved by NERSA.

1.2. Handling of Non-compliances and Deviations

- (1) Amendments, derogations or exemptions shall be processed in accordance with the requirements of the RSA Grid Code, as amended.

1.3. Capability to provide ancillary services

- (1) Categories B2 and C shall have a capability to provide ancillary services that will be concluded in line with section 4 of the System Operation Code

2. Objectives

- (1) The primary objective of this grid connection code is to specify minimum technical and design grid connection requirements for Battery Energy Storage Facilities (*BESF*) connected to or seeking connection to the South African electricity transmission system (*TS*) or distribution system (*DS*).
- (2) This document shall be used together with other applicable requirements of the code (i.e. the South African Grid Code, the Distribution Code and the Scheduling and Dispatch Rules), as compliance criteria for *BESF* connected to the *TS* or the *DS*.

3. Scope

- (1) The grid connection requirements in this code shall apply to all *BESF* connected or seeking connection to the *TS* or *DS*, *the SO*, as well as to the respective electrical *Network Service Providers (NSPs)*.
- (2) The *BESF* can be connected as a standalone facility or as multiple units operating as modular parallel units forming a large integrated facility. In the latter the facility category is based on the combined nominal active power of the connected units

- (3) The *NSP* determines the *POC* of the *BESF*
- (4) Unless otherwise stated, the requirements in this grid connection code shall apply equally to all *BESF* categories in charging or discharging mode.
- (5) Where there has been a replacement of or a major modification as defined in section 4 to an existing *BESF*, the *BESF* shall be required to demonstrate compliance to these requirements before being allowed to operate commercially.
- (6) Compliance with this grid connection code shall be applicable to the *BESF* depending on its nominal AC active power and, where indicated, the nominal voltage at the *POC*. Accordingly, *BESF* are grouped into the following categories shown in Table 1:

Table 1: *BESF* categories

Category	Rated power of <i>BESF</i>		
A	>0	to	< 1 MW
A1	>0	to	≤ 13.8 kW
A2	>13.8 kW	to	<100 kW
A3	≥100 kW	to	<1 MW
B	≥1 MW	to	<20 MW
B1	≥1 MW	to	<5 MW
B2	≥5 MW	to	<20 MW
C	≥20 MW	-	-
Note: For a category A <i>BESF</i> connected to multi-phase supplies (two- or three-phase connection at the <i>POC</i>), the difference in installed capacity between phases may not exceed 4.6 kW per phase			

- (7) The *NSP* shall supply the *BESF* Owner with a reasonable detail of their *TS* or *DS* that is sufficient to allow an accurate analysis of the interaction between the *BESF* and the *NIPS*, including other generation facilities. Network technical data shall be provided in line with the requirements of Appendix 4, Provision of Technical Grid Data to *BESF* Owners

Note: A UPS system (uninterruptible power supply with batteries) is not defined as a *BESF* and is therefore not covered by this code, as the function of a UPS is to maintain power supply locally in an installation or in a part of an installation in case of public electricity supply system disturbances or failure.

4. Definitions and Abbreviations

- (1) Unless otherwise indicated, words and terminology in this document shall have the same meaning as those in the *codes*. The following definitions and abbreviations are used in this document.

4.1. Definitions and abbreviations with explanation

90% error band

Error band calculated in such a way that 90% of sample-by-sample errors are within this band. 10% of samples may have errors higher than the band.

Active Power (MW) Curtailment Set-point

The limit set by the *SO*, *NSP* or their agent for the amount of active power that the *BESF* is permitted to generate or absorb. This instruction may be issued manually or automatically via a tele-control facility. The manner of applying the limitation shall be agreed between the parties.

Ancillary services

Services supplied to the *NTC* by generators, *NSPs* or end-use customers, necessary for the reliable and secure transport of power from generators to *NSPs* and other customers, as defined in the System Operation Code, section 4.

Available Active Power ($P_{\text{available}}$)

The amount of active power (MW or kW), measured at the *POC*, that the *BESF* could deliver or absorb based on charging capacity.

Average absolute error

Average of absolute values of errors over a certain period.

Average error

Average of errors over a certain period.

Battery Energy Storage Facility (*BESF*)

Battery Energy Storage Facility comprises batteries, chargers, power converters and related equipment connected to a single point of connection (*POC*) on the *NIPS* for the purpose of storing electrical energy in the batteries during the charging process and discharging the stored electrical energy when required.

Battery Energy Storage Facility (*BESF*) Controller

A set of control functions that make it possible to control the *BESF* connected to a single point of connection (*POC*). The set of control functions shall form a part of the *BESF*.

Battery Energy Storage Facility (*BESF*) Owner

Means a legal entity that is licensed or registered to develop and operate a *BESF*.

Charging Capacity

The maximum amount of energy imported and stored by the *BESF* during charging from minimum discharge level to maximum charged level at standard testing conditions and rated power, expressed in kWh.

Codes

The Distribution Code, the Transmission Grid Code or any other Code approved by *NERSA*.

Communication Gateway Equipment

The equipment forming the link between the communications equipment (e.g. the Data Terminal Equipment (DTE, or sometimes called a Remote Terminal Unit (RTU)) of the System Operator's SCADA / Energy Management System) and the *participant's* plant.

Connection Agreement

An agreement detailing the conditions under which the *NSP* intends to connect the *customer*. This agreement specifies the conditions applicable to an end use customer or any other connection such as a *BESF*. This agreement is entered into after the offer to connect is accepted by the *customer*.

Frequency Control band

The frequency control band is the frequency band within the frequency thresholds 49.5 Hz to 50.5 Hz and where the active power is controlled with the intention to maintain the 50Hz system frequency.

Curtailed Active Power

The amount of Active Power that the *BESF* is permitted to deliver or absorb by the *SO*, *NSP* or their agent subject to network or system constraints.

Customers

A person who purchases electricity or a service relating to the supply of electricity

Dead band

Interval used intentionally to make a control function unresponsive;

Distribution System (DS)

The network infrastructure operating at nominal voltages of 132kV or less

Electromagnetic Compatibility

Electromagnetic compatibility (EMC) requirements as per standard referenced.

Equivalent Network

This is a reduced network consisting of a source infeed/s, network boundary bus(es), interconnecting transformer(s) and surrounding network that represents the behaviour of the external system as seen from its network boundary bus.

External network

The rest of the network which is represented by the network boundary bus and source infeed(s).

Fault level(s)

Known as Short circuit level(s) – Provides information on the strength of the network and equivalent source impedance under defined conditions.

Frequency

The number of oscillations per second on the AC waveform.

Frequency control

Automatic active power regulation in response to a measured deviation of system frequency beyond pre-set thresholds, in order to maintain stable system frequency of the *NIPS*

Frequency droop

A frequency droop, *S*, means the ratio of a change of frequency to the resulting change in active power output, expressed in percentage terms. The change in frequency is expressed as a ratio to nominal frequency and the change in active power expressed as a ratio to nominal active power of *BESF*.

$$S = 100\% \times \frac{\Delta_f}{f_n} \times \frac{P_{ref}}{\Delta_P} \quad [\%]$$

S: Droop in %

Δ_f : System frequency change, where the *BESF* must provide a frequency response

f_n : Nominal system frequency

P_{ref} : Nominal power of the *BESF*

Δ_P : The change in active power performed by the *BESF*

Frequency sensitive mode (FSM)

‘Frequency sensitive mode’ (FSM) means the operating mode of a *BESF* in which the active power output changes in response to a change in system frequency, in such a way that it assists with the recovery to a target frequency.

High voltage (HV)

The set of nominal voltage levels greater than 33 kV up to and including 132 kV.

Harmonic Distortion

Harmonic distortion is defined as the ratio of harmonics to fundamental frequency component when a (theoretically) pure sinewave is reconstructed.

Interconnecting Transformers

This is the network transformer(s) that connect the surrounding network to the network boundary bus representing the external network.

Interconnected power system (IPS)

The *IPS* consists of

- the *TS*
- assets *connected to the TS* and belonging to the *NTC*
- *power stations* connected to the *TS*
- international interconnectors
- the *control area* for which the *System Operator* is responsible.

The IPS definition is not linked to specific assets, but includes those components of the electrical network that have a measurable influence, at transmission level, on each other as they are operating as one power system.

Loss of mains detection (LOM)

Detection scheme that identifies unintended unit island operation and activates required *BESF* action.

Low voltage (LV)

Nominal voltage levels up to and including 1 kV.

Lumped Generation

This refers to the aggregated sum of individual generation units connected at a specific busbar.

Lumped Load

This refers to the aggregated sum of individual loads connected at a specific busbar.

Major modification

A major modification is assessed as being when an existing *BESF* is modified to such an extent that already compliance tested specifications, capabilities and characteristics change.

Depending on the nature and impact of the modification an updated or new operating agreement is necessary.

An updated or new grid connection agreement is necessary when;

- Removing a functionality or adding a functionality i.e. change in the BESF specification
- Modifying/updating a functionality and as a result of this – change of unit specified performance
- Capacity is removed or added
-

The compliance tested capabilities and performance characteristic of the grid connected *BESF* must be identical with values included in the operating agreement throughout the lifetime of the *BESF*.

Maximum short circuit level

Maximum short circuit level must be calculated using IEC 60909 method and network configuration that will provide the highest value at a particular busbar i.e. with all fault current contributing elements in service

Maximum voltages (U_{max})

Maximum continuous operating voltage.

Medium voltage (MV)

The set of nominal voltage levels greater than 1 kV up to and including 33 kV.

Minimum short circuit level

Minimum short circuit level must be calculated using complete fault current calculation method and network configuration that will provide the lowest value at a particular busbar. This is achieved by modelling minimum generation in service and all possible outages of fault current contributing elements (lines, transformers, etc.) provided that the network is still within an operable range.

Minimum voltages (U_{min})

Minimum continuous operating voltage.

National Energy Regulator of South Africa (NERSA)

The legal entity established in terms of the National Energy Regulator Act, 2004 (Act 40 of 2004).

NERSA replaces the National Electricity Regulator (NER) which was established in terms of the Electricity Act, 1987 (Act No.41 of 1987).).

National Interconnected Power Systems (NIPS)

The electrical network comprising components that have a measurable influence on each other as they are operating as one system, this includes:

- the *TS*;
- the *DS*;
- assets connected to the *TS* and *DS*;
- *generators* connected to the *TS* and *DS*;
- international *interconnectors*;
- the *control area* for which the *SO* is responsible.

National Transmission Company (NTC)

The South African legal entity licensed to execute the national *transmission* responsibility. It consists of a *System Operator* and a national *transmission network service provider*.

Network Boundary

Network boundary is one voltage level above the POC, e.g. if POC is 132 kV, then information shall be at 400 kV, 275 kV or 220 kV, It is typically one voltage level higher than POC.

Network Service Provider (NSP)

A legal entity that is licensed to provide network services through the ownership and maintenance of an electricity network

Nominal power delivered by the Battery Energy Storage Facility (P_{nd})

Nominal (Maximum) active power output which the *BESF* is designed to continuously deliver measured at *POC*.

Nominal power absorbed by the Battery Energy Storage Facility (P_{na})

Nominal (Maximum) active power input which the *BESF* is designed to continuously absorb at *POC*.

Nominal voltage

The utility voltage for which a network is defined and to which operational measurements are referred.

Normal Operating Conditions

An operating condition where the system frequency, voltage and equipment loading are within their statutory, contractual and/or design limits and no network component on the relevant part of the *DS* or *TS* is out of service due to an outage.

Participants

As defined in the *Code*,

Point of Common Coupling (PCC)

The electrical node, normally a busbar, in a *transmission or distribution substation* where different feeds to *customers* are or may be connected together.

Point of Communication (PCOM)

The point in an energy storage facility where the data communication properties are made available and verified between *BESF* and the *NSP/ SO*.

Point of Connection (POC)

The electrical node on a distribution or transmission system where a customer's assets are physically connected to the NSPs assets through a bay.

Power Quality (PQ)

Characteristics of the electricity at a given point on an electrical system, evaluated against a set of reference technical parameters. These characteristics include:

- voltage or current quality, i.e. regulation (magnitude), harmonic distortions, flicker, unbalance;
- voltage events, i.e. voltage dips, voltage swells, voltage transients;
- (supply) interruptions;
- frequency of supply.

RoCoF

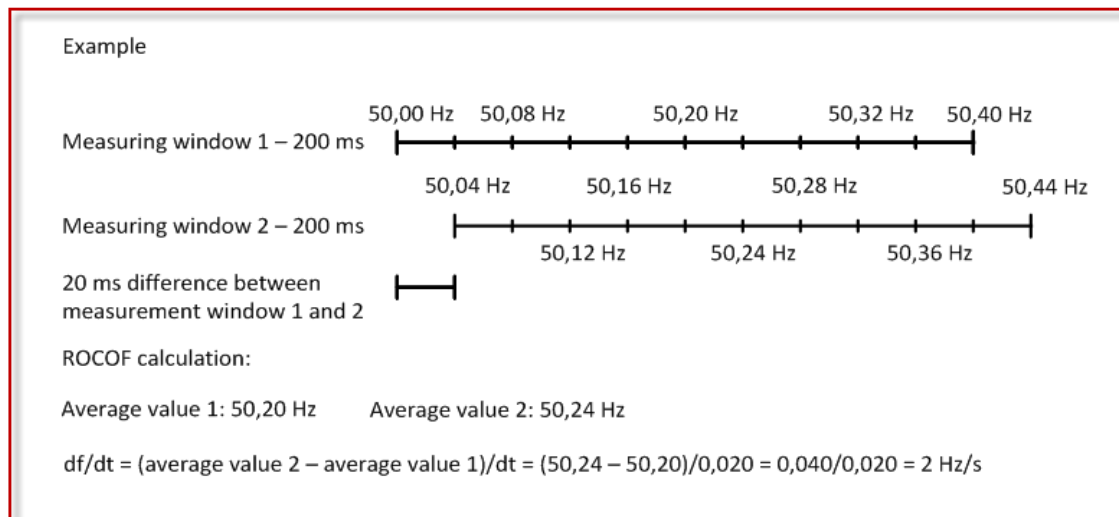
Rate of Change of Frequency (ROCOF) is the term for frequency change as a function of time.

The RoCoF, is calculated as follows:

- The frequency measurements used for calculation of the frequency change is based on a 200 ms measuring window whereof the average value is calculated.
- The frequency shall be continuously measured, thus a new average frequency value can be calculated every 20 ms.
- RoCoF [Hz/s] shall be calculated as the difference between the most recent frequency average calculation (Average of measuring window 2) and the average value calculated 20 ms ago (Average of measuring window 1)

$$\text{RoCoF} = df/dt = (\text{average 2} - \text{average 1})/0,020 \text{ [Hz/s]}$$

Note:



Sample-by-sample error

Percentage error calculated as a ratio of difference between recorded and simulated values of synchronized samples referred to maximum peak value of recorded voltage or current in any phase.

State of Charge (SOC)

The degree to which a BESF is charged relative to the charging capacity that can be stored, expressed as a percentage.

Surrounding network

This is the network at the same voltage level as the POC (e.g. surrounding 132kV network if POC is at 132kV or 33kV network if POC is at 33kV). For extensive surrounding networks, parts of the network may be represented by lumped sources and/or lumped loads.

System Operator (SO)

The legal entity licensed to be responsible for short-term reliability of the IPS, which is in charge of controlling and operating the TS and dispatching generation (or balancing the supply and demand) in real time.

Total Harmonic Distortion

$$\text{Total harmonic distortion: } \text{THD}_I(\%) = \sqrt{\sum_{n=2}^{50} I_n^2}$$

Where I_n is the per cent RMS value of the nth harmonic current component

Transmission Network Service Provider (TNSP)

A legal entity that is licensed to own and maintain a network on the TS.

Transmission System (TS)

The TS consists of all lines and *substation* equipment where the nominal voltage is above 132 kV.

All other equipment operating at lower voltages are either part of the *distribution system* or classified as *transmission transformation equipment*.

Voltage Droop

A voltage droop means the ratio of a change of voltage to the resulting change in reactive power output, expressed in percentage terms. The change in voltage is expressed as a ratio to operational voltage and the change in reactive power expressed as a ratio to nominal reactive power capacity; i.e. from max reactive power import to max reactive power export.

Voltage Fault Ride Through (VFRT) Capability

Capability of the *BESF* to stay connected to the network and keep operating following voltage dips or surges caused by short-circuits faults or disturbances on any or all phases in the *TS* or *DS*.

Voltage Quality

Subset of *power quality* referring to steady-state voltage quality, i.e. voltage regulation (magnitude), voltage harmonics, voltage flicker, voltage unbalance, voltage dips. The current drawn from or injected into the *POC* is the driving factor for voltage quality deviations.

5. Tolerance of Frequency and Voltage Deviations

- (1) The *BESF* shall be able to withstand frequency and voltage deviations at the *POC* under normal and abnormal operating conditions described in this grid connection code while reducing the active power as little as possible.
- (2) The *BESF* shall be able to support network frequency and voltage stability in line with the requirements of this grid connection code.
- (3) Normal operating conditions and abnormal operating conditions are described in section 5.1 and section 5.2, respectively.

5.1. Normal Operating Conditions

Unless otherwise stated, requirements in this section shall apply to all *categories* of *BESF*.

- (1) *BESF* of *Category A* shall be designed to be capable of operating continuously within the voltage range of -15% to +10% of the nominal voltage at the *POC*.
- (2) *BESF* of *Category B1, B2 and C* shall be designed to be capable of operating continuously within the *POC* voltage range specified by *Umin* and *Umax* as shown in Table 2 below, measured at the *POC*.

Table 2: Minimum and maximum operating voltages at POC

Nominal voltage, U_n [kV]	U_{min} [pu]	U_{max} [pu]
765	0.95	1.05
400	0.95	1.05
275	0.95	1.05
220	0.95	1.05
132	0.90	1.0985
88	0.90	1.0985
66	0.90	1.0985
44	0.90	1.08
33	0.90	1.08
22	0.90	1.08
11	0.90	1.08

(3) The nominal frequency of the *National Integrated Power System (NIPS)* is 50 Hz and is normally controlled within the limits as defined in the *Grid Code*. The *BESF* shall be designed to be capable of operating for the minimum operating range illustrated in Figure 1(during a system frequency disturbance).

(4) When the frequency on the *NIPS* is higher than 51.5 Hz for longer than 4 seconds, the *BESF* is allowed to disconnect from the grid.

(5) When the frequency on the *NIPS* is less than 47.0 Hz for longer than 200 ms, the *BESF* is allowed to disconnect from the grid.

(6) The *BESF* shall remain connected to the *NIPS* during rate of change of frequency (RoCoF) of values up to and including ± 2.5 Hz per second, measured with a 200 ms moving window, provided that the system frequency is still within the minimum operating range indicated in Figure 1 below.

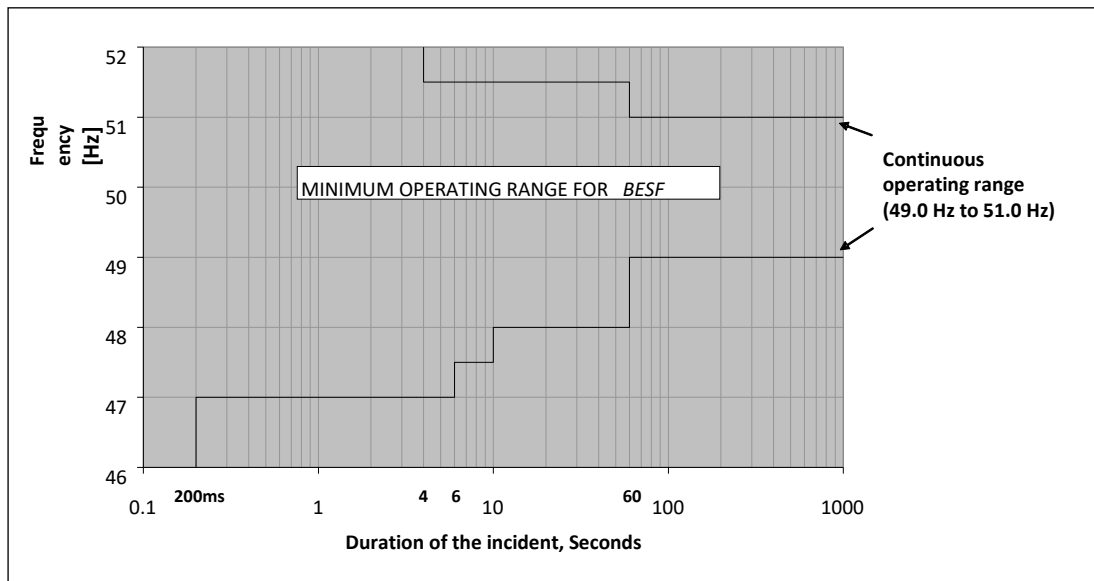


Figure 1: Minimum frequency operating range for a BESF (during a system frequency disturbance)

5.2. Synchronizing to the NIPS

(1) *BESF of Category A* shall only be allowed to connect to the *NIPS*, at the earliest, 60 seconds after the following conditions have been satisfied:

- (a) the voltage at the *POC* is in the range -15 % to +10 % of the *nominal voltage*,
- (b) *frequency* in the *NIPS* is within the range of 49.0 Hz and 50.2 Hz, or otherwise as agreed with the *SO*.

(2) *BESF of Category B1, B2 and C* shall only be allowed to connect to the *NIPS*, at the earliest, 3 seconds after the following conditions have been satisfied:

- (a) the voltage at the *POC* is within *U_{max}* and *U_{min}*, as specified in 2, around the *nominal voltage*,
- (b) *frequency* in the *NIPS* is within the range of 49.0 Hz and 50.2 Hz, or otherwise as agreed with the *SO*.

5.3. Abnormal Operating Conditions

5.3.1. Tolerance to phase jump

(1) The *BESF* shall be designed to withstand sudden phase jumps not related to system failure of up to 20° at the *POC* without disconnecting or reducing its active power output. The *BESF* shall after a settling period resume normal production not later than 5 sec after the operating conditions in the *POC* have reverted to the *normal operating conditions*.

5.3.2. Tolerance to sudden voltage drops and peaks

(a) *BESF of Category A1 and A2*

(1) *BESF of Categories A1 and A2* shall be designed to withstand and fulfil, at the *POC*, voltage ride through conditions illustrated in Figure below.

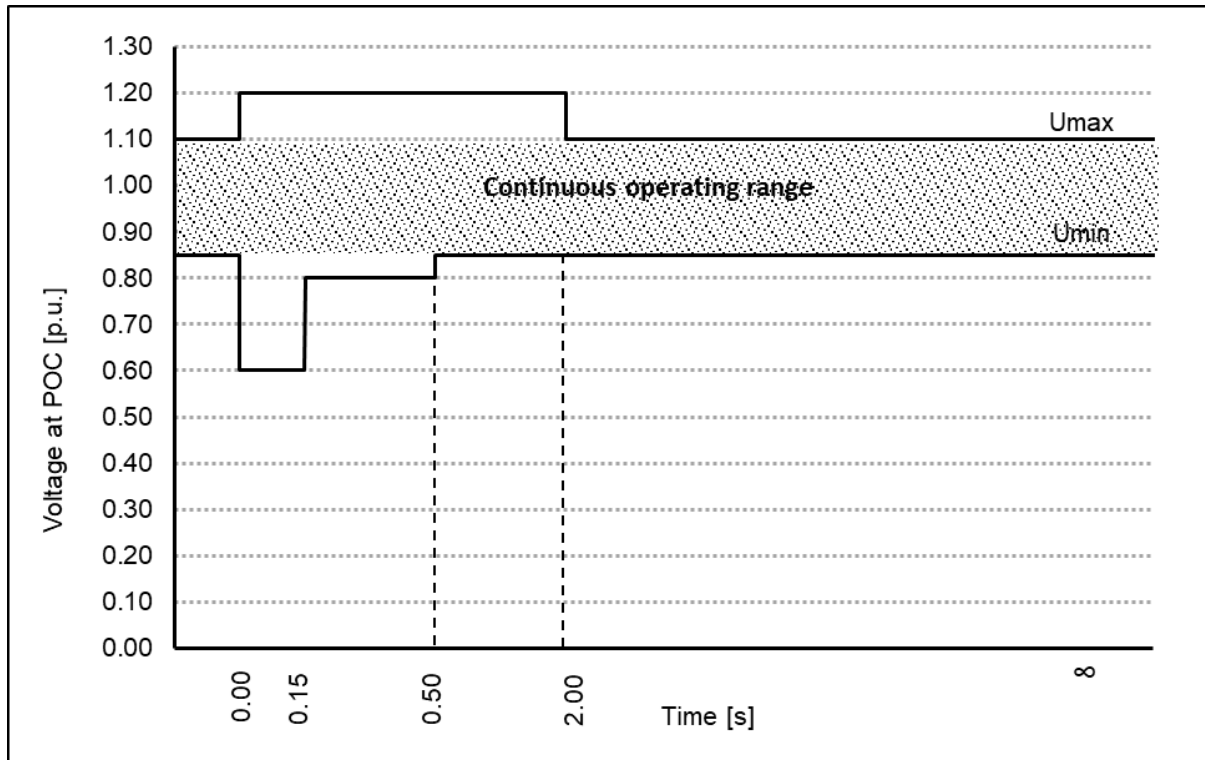


Figure 2 - Voltage Ride Through Capability for the BESF of Category A1 and A2.

(b) BESF of Categories A3, B1, B2 and C

- (1) The BESF shall be designed to withstand voltage drops and peaks, as illustrated in Figure 3. During the fault the BESF must supply or absorb reactive current as illustrated in Figure 3 without disconnecting.
- (2) The BESF shall be able to withstand voltage drops to zero, measured at the POC, for a minimum period of 0.150 seconds without disconnecting, as shown in Figure 3.
- (3) The BESF of category C shall be able to withstand voltage peaks up to 120% of the nominal voltage, measured at the POC, for a minimum period of 2 seconds without disconnecting, as shown in Figure 3.
- (4) Figure 3 shall apply to all types of faults (symmetrical and asymmetrical i.e. one-, two- or three-phase faults) and the bold line shall represent the minimum voltage of all the phases.

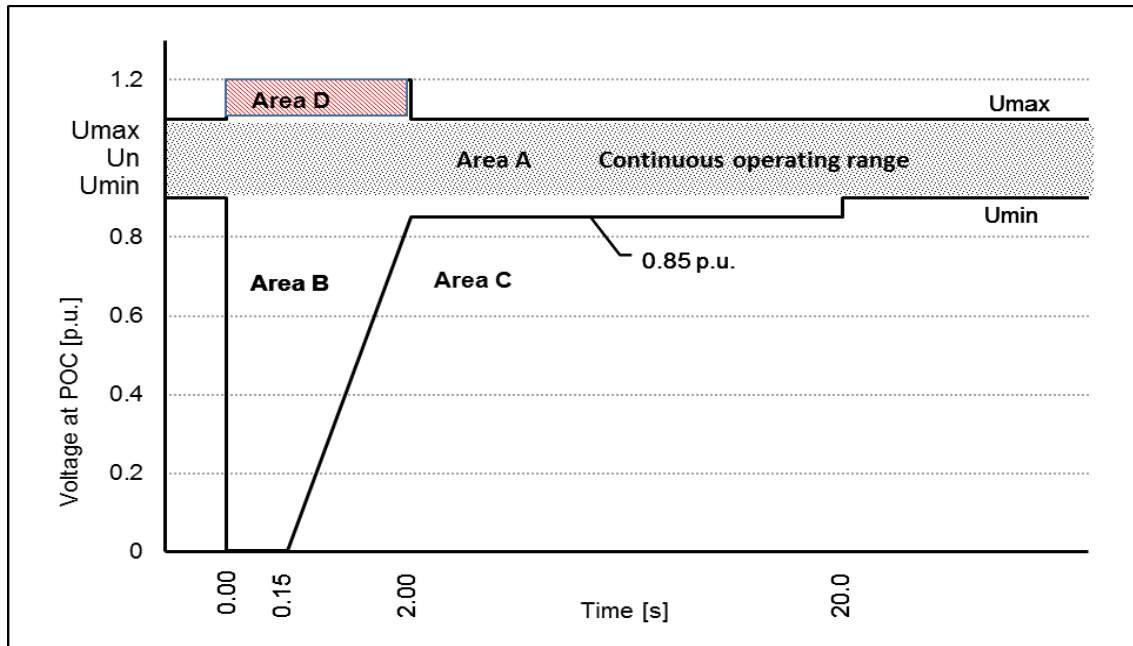


Figure 3 - Voltage Ride through Capability for the BESF of Category A3, B1, B2 and C.

- (5) If the voltage (U) reverts to area A during a fault sequence, subsequent voltage drops shall be regarded as a new fault condition. If several successive fault sequences occur within area B and evolve into area C, disconnection is allowed.
- (6) In the area C disconnection of the *BESF* is allowed but is not mandatory.
- (7) In connection with symmetrical fault sequences in areas B and D of, the *BESF* shall have the capability of controlling the positive sequence of the reactive current, as illustrated in Figure 3. The following requirements shall be complied with:
- (a) **Area A:** The *BESF* shall stay connected to the network and uphold normal production.
 - (b) **Area B:** The *BESF* shall stay connected to the network and in addition:
 - (i) *BESF* of category A shall not inject any reactive current into the network;
 - (ii) *BESF* of category B1, B2 and category C shall provide maximum voltage support by supplying a controlled amount of additional reactive current as shown in Figure 3 so as to ensure that the *BESF* contributes to stabilizing the voltage within the design framework offered by the *BESF*.
 - (iii) *BESF* of category B1, B2 and category C shall be able to disable reactive current support functionality at the request of the SO or the local network operator.
 - (c) **Area D:** The *BESF* shall stay connected to the network and provide maximum voltage support by absorbing a controlled amount of reactive current so as to ensure that the *BESF* helps to stabilize the voltage within the design capability offered by the *BESF*. See Figure 3.

- (d) **Area E (in Figure 4)** :Once the voltage at the *POC* is below 20%, the *BESF* shall continue to supply reactive current within its technical design limitations so as to ensure that the *BESF* helps to stabilize the voltage. Disconnection is only allowed after conditions of Figure 3 have been fulfilled.
- (8) Control shall follow Figure 4 so that the reactive current follows the control characteristic with a tolerance of $\pm 20\%$ after 60 ms.
- (9) The supply of reactive power has first priority in area B, while the supply of active power has second priority. Active power shall be maintained during voltage drops, but a reduction in active power within the *BESF* design specifications is required in proportion to voltage drop for voltages below 85 %.
- (10) Upon clearance of fault, each *BESF* shall restore active power production to at least 90 % of the level available immediately prior to the fault within 1 second.

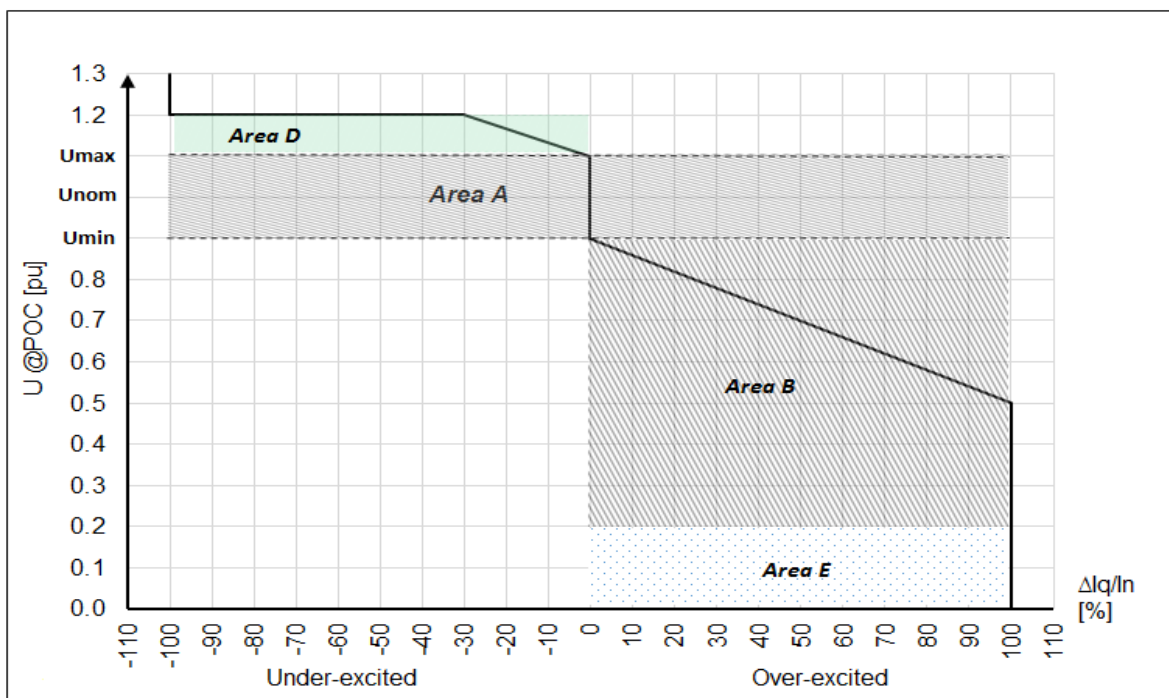


Figure 4 - Requirements for Reactive Current, I_Q , during a voltage failure.

6. Frequency Response

- (1) In case of frequency deviations in the *NIPS*, all *BESF* categories shall be capable of providing the specified frequency response in order to stabilize the grid system *frequency*.
- (2) The type of frequency response applicable to all categories is indicated in table 3 below.

Table 3: Frequency response requirement for *BESF*

	Category A	Category B	Category C
Low frequency, mandatory response)	Optional	Optional	Required
High frequency, mandatory response)	Required	Required	Required
(Normal/ contracted frequency response)	Optional	Required	Required

- (3) The metering accuracy for the grid system *frequency* shall be ± 10 mHz or better.
- (4) It shall be possible to set the frequencies to the default settings for $f_{\min}$, $f_{\max}$, f_1 , to f_6 shall be as shown in Table 4, unless otherwise agreed upon between the *SO* and the *BESF*.

Table 4 - Frequency Default Settings

Parameter	$f_{\min}$	$f_{\max}$	F_0	f_1	f_2	f_3	f_4	f_5	f_6
Frequency [Hz]	47,00	52,00	50,00	47,50	49,50	49,85	50,15	50,50	51,50

- (5) The *BESF* shall according to *BESF* category requirement, be equipped with the frequency control *droop* settings.
- (6) The default settings for *Droop 1* to *Droop 4* shall be as shown in Table 5 below, unless otherwise agreed upon between the *SO* and the *BESF*. The actual *droop* setting shall be as agreed with the *SO*.

Table 5 - Droop Default Settings

Parameter	Droop 1	Droop 2	Droop 3	Droop 4
% of nominal active power capacity	10 %	4 %	4 %	10 %

6.1. Frequency response – Low frequency range

- (1) This section shall apply to *BESFs* of category *C* only.
- (2) During low frequency (below f_2) operating conditions, *BESF* shall be able to provide mandatory, instantaneous and autonomous *active power* response in order to support the frequency in accordance with *droop 1* in Figure 5 below. The metering accuracy for the grid frequency shall be ± 10 mHz or better.
- (3) The action taken by the *BESF* will depend on whether it is in charging or discharging mode. The *BESF* shall respond as depicted in Figure 6 as follows to a decreasing system frequency:
 - a) While in charging mode, if frequency drops below f_2 , the *BESF* shall reduce demand and change to discharge mode.
 - b) if the system frequency still drops to below f_1 the *BESF* shall stay connected at least until $f_{\min}$ or until the protection is disconnecting the *BESF*.
 - c) While in discharging mode, no decrease in output shall occur. The *BESF* shall continue discharging and respond according to droop 1 as per Figure 5.
 - d) The low frequency response can be interrupted when the system frequency is above f_2 or if the *BESF* discharge level reaches an agreed level according to the connection agreement.

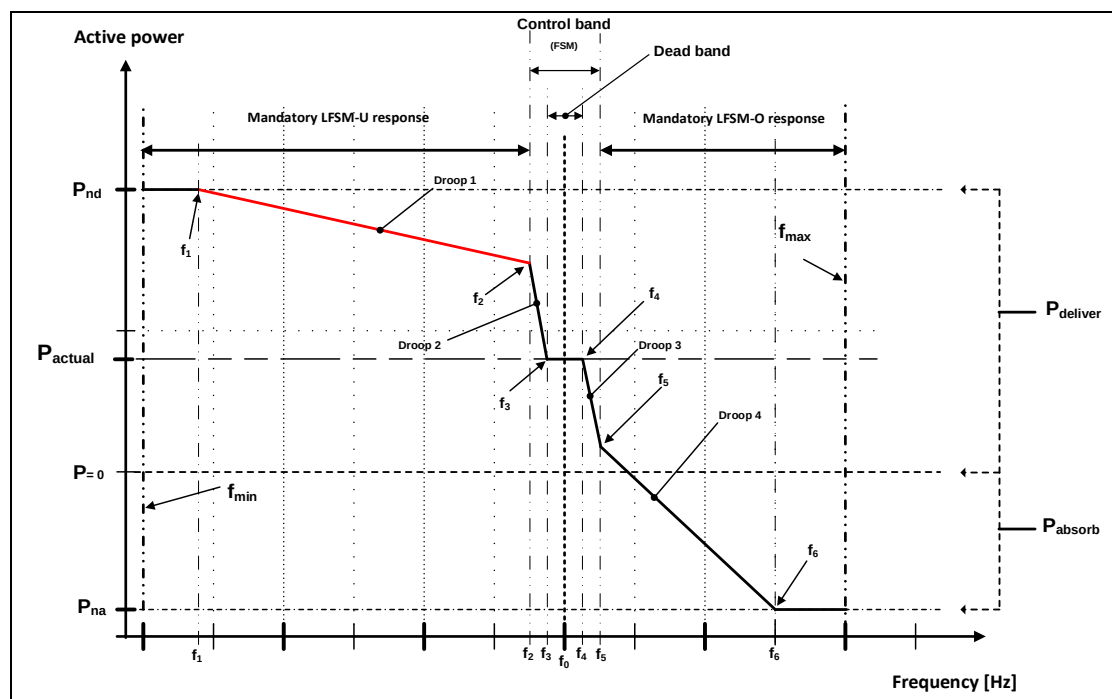


Figure 5 - Power response during low frequency

6.2. Frequency response- High frequency range

(1) This section shall apply to *BESFs* of all categories.

(2) During high frequency (above f_5) operating conditions, *BESF* shall be able to provide mandatory, instantaneous and autonomous active power response in order to stabilize the system frequency in accordance with droop 4 in Figure 6 . The action taken by the *BESF* will depend on whether it is in discharging or charging mode. The *BESF* shall respond as follows to reduce system frequency:

- While in discharging mode, if system frequency increases above f_5 , the *BESF* shall gradually reduce output and change to charging mode if frequency is still increasing to above f_6 in accordance with Figure 6.
- If the system frequency increase above f_6 the *BESF* shall remain connected at least until $f_{\max}$ or until the protection is disconnecting the *BESF*.
- While in charging mode, no decrease in demand shall occur. The *BESF* shall continue charging and respond according to droop 4 as per Figure 6.
- The high frequency response can be interrupted when the system frequency is below f_5 or if the *BESF* charging level reach an agreed level according the connection agreement. (e.g.100%)

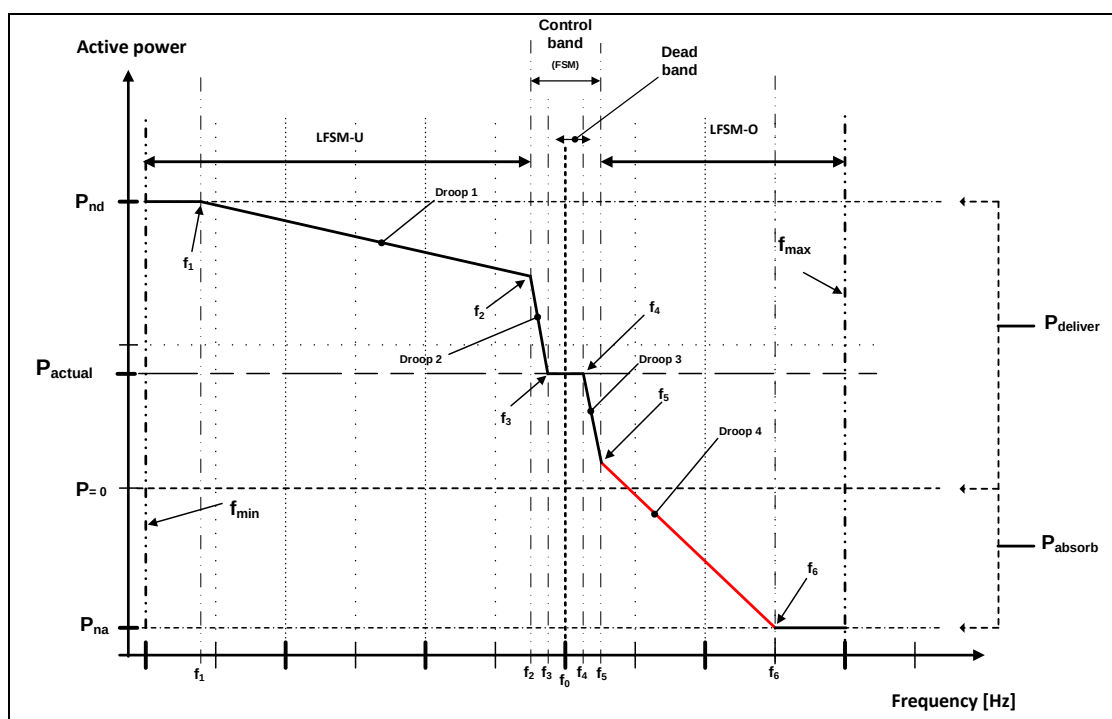


Figure 6 - Power response during high frequency.

6.3. Frequency response – Normal/ contracted frequency range

- (1) *BESFs* of category *B1*, *B2* and *C*, shall be capable of providing power-frequency response as illustrated with droop 2 and droop 3 in Figure 7.
- (2) The purpose of the frequency points f_2 and f_5 is to form the *control band (FSM)* for contracted ancillary services. In addition, the *frequency* points f_2 and f_5 identify the *frequency* threshold below and above the control band where the autonomous low and high *frequency* response is required.
- (3) The purpose of the *frequency* points f_3 and f_4 is to form the dead band within the *control band*
- (4) The *SO* shall decide and advise the *BESF* (directly or through its agent) on the *droop* settings required to perform control between the various *frequency* points.
- (5) It shall be possible to activate and deactivate the *frequency* response control function between $f_{\min}$ to $f_{\max}$.
- (6) If a *frequency* control parameter ($f_1 - f_6$) is to be changed as instructed by *SO*, such change shall be commenced within two seconds and completed no later than 10 seconds after receipt of an order to change the parameter.
- (7) The accuracy of the control performed (i.e. change in active power output) and of setpoints shall not deviate by more than $\pm 2\%$ of the setpoint value or by $\pm 0.5\%$ of the rated power (P_{nl}), depending on which yields the highest tolerance.

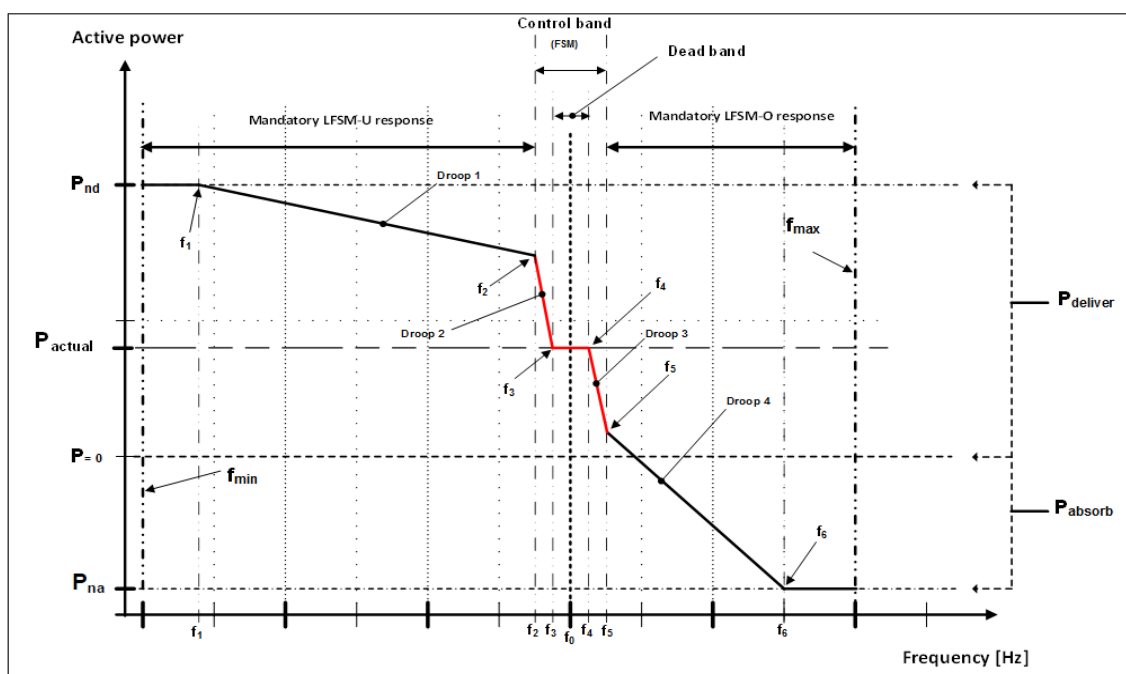


Figure 7 –Power response during normal/ contracted frequency range

7. Active Power Constraint Functions

- (1) This section shall apply to BESF of categories A3, B1, B2 & C
- (2) A *BESF* must be equipped with control functions capable of controlling the active power at the *POC* in discharging and charging modes.
- (3) For system security reasons it may be necessary for the SO, NSP or their agent to curtail the BESF active power output or input whilst is in discharging or charging mode.
- (4) The *BESF* shall in discharging or charging modes be capable of:
 - Operating the *BESF* at a reduced level if active power has been curtailed by the SO, NSP or their agent for network or system security reasons,
 - receiving a telemetered MW Curtailment set-point sent from the SO, NSP or their agent. If another operator is implementing power curtailment, this shall be in agreement with all the parties involved.
- (5) The *BESF* shall be equipped with constraint functions, i.e. supplementary active power control functions. The constraint functions are used to avoid imbalances in the *NIPS* or overloading of the *TS* and *DS* in connection with the reconfiguration of the *TS* and *DS* in critical or unstable situations or the like.
- (6) Activation of the active power constraint functions shall be agreed with the *SO* or *NSP*. The required constraint functions are as follows:
 - (a) Absolute production constraint
 - (b) Power gradient constraint.

7.1. Absolute Production Constraint

- (1) An Absolute Production Constraint is used to constrain the output of active power from the *BESF* to a predefined power MW limit at the *POC*.
- (2) If the setpoint for the Absolute Production Constraint is to be changed, the *BESF* shall commence such change within two seconds and the change shall be completed not later than 30 seconds after receipt of an order to change the setpoint.
- (3) The accuracy of the control performed and of the setpoint shall not deviate by more than $\pm 2\%$ of the setpoint value or by $\pm 0.5\%$ of the rated power, depending on which yields the highest tolerance, average over a 1-minute interval.

7.2. Power Gradient Constraint

- (1) A Power Gradient Constraint (ramp rate) is used to limit the maximum speed by which the active power can be changed in the event of changes in power or in the set points for a *BESF*.
- (2) It shall be possible to set the ramp rate limit to any value between 1 to 20 % of P_{nd} or P_{na} per minute for both upward and downwards ramp rates. The actual ramp rate, for both upward and downwards regulation must not exceed 60 MW per minute unless specified and agreed by the SO.
- (3) The accuracy of the control performed and of the setpoint shall not deviate by more than $\pm 2\%$ of the setpoint value or by $\pm 0.5\%$ of the rated power, depending on which yields the highest tolerance, average over a 1-minute interval.

8. Reactive power capability

8.1. Category A BESF

- (1) The *BESF* of category A1 & A2 shall be designed with the capability to operate in a power factor control mode and must, by default, follow a power factor of 1.00 measured at the *POC*, unless otherwise specified by the *NSP* or the *SO*.
- (2) The *BESF* of category A3 shall be designed with the capability to supply and absorb rated active power, P_{nd} and P_{na} [MW] for power factors ranging between 0.975 lagging and 0.975 leading, measured at the *POC* and available from 20 % to 100 % of rated power (P_{nd}/P_{na}).
- (3) When the *BESF* of category A is disconnected or is not delivering or absorbing any active power, no compensation of reactive power is required from the *BESF*.

8.2. Category B and C BESF

- (1) *BESF* of category B and C shall be able to generate and consume reactive power (Q) at the *POC* continuously within the operating range limited by its maximum rated capacity in production mode and demand mode shown in Figure 8, where Q_{min} and Q_{max} are voltage dependent as defined by Figure 9. The voltage ranges $U_{min} - U_{max}$ are defined in section 5.1.
- (2) For *BESF* of category B, $Q_{min} = -0.228 \cdot P_{na}/P_{nd}$ and $Q_{max} = 0.228 \cdot P_{na}/P_{nd}$. At rated active power, P_{nd} and P_{na} [MW], this corresponds to a power factor range of 0.975 leading to 0.975 lagging.
- (3) For *BESF* of category C, $Q_{min} = -0.33 \cdot P_{na}/P_{nd}$ and $Q_{max} = 0.33 \cdot P_{na}/P_{nd}$. At rated active power, P_{nd} and P_{na} [MW], this corresponds to a power factor range of 0.95 leading to 0.95 lagging.

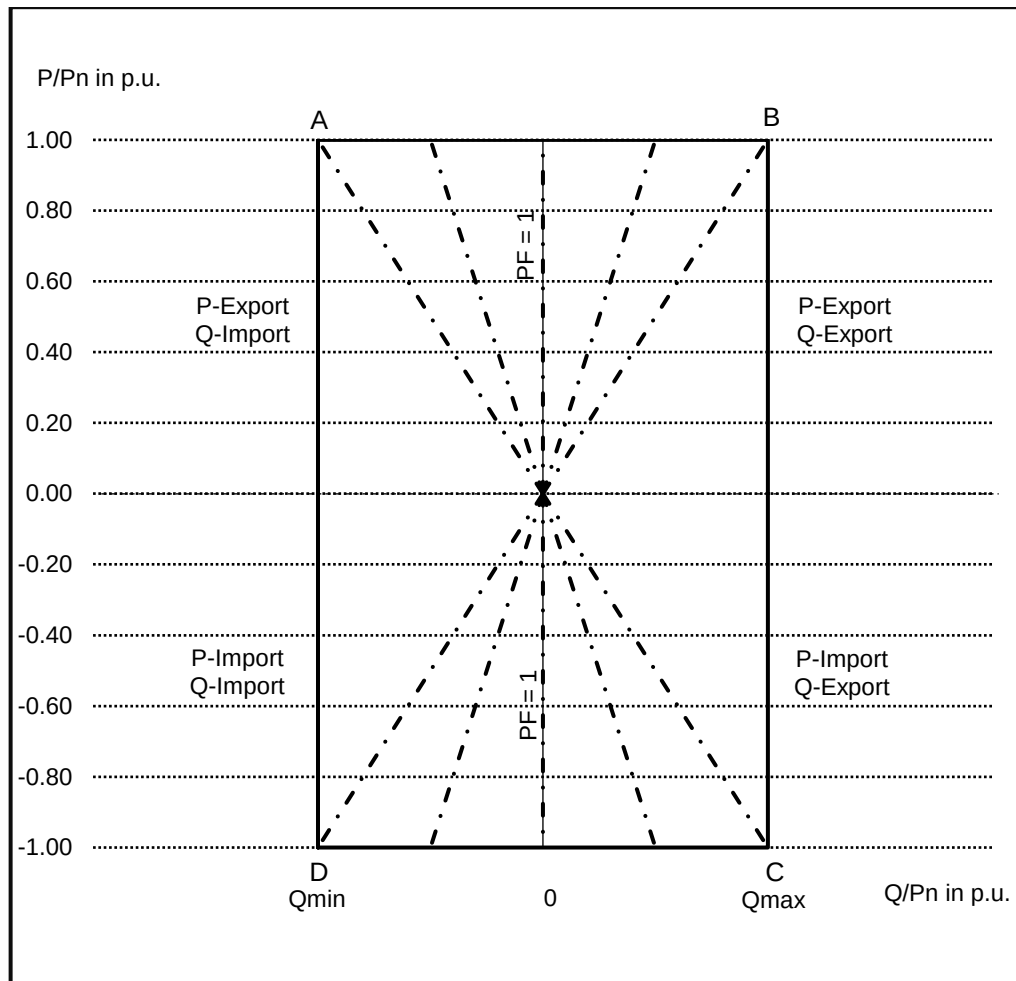


Figure 8: Reactive power capability for BESF of category B and C at the POC

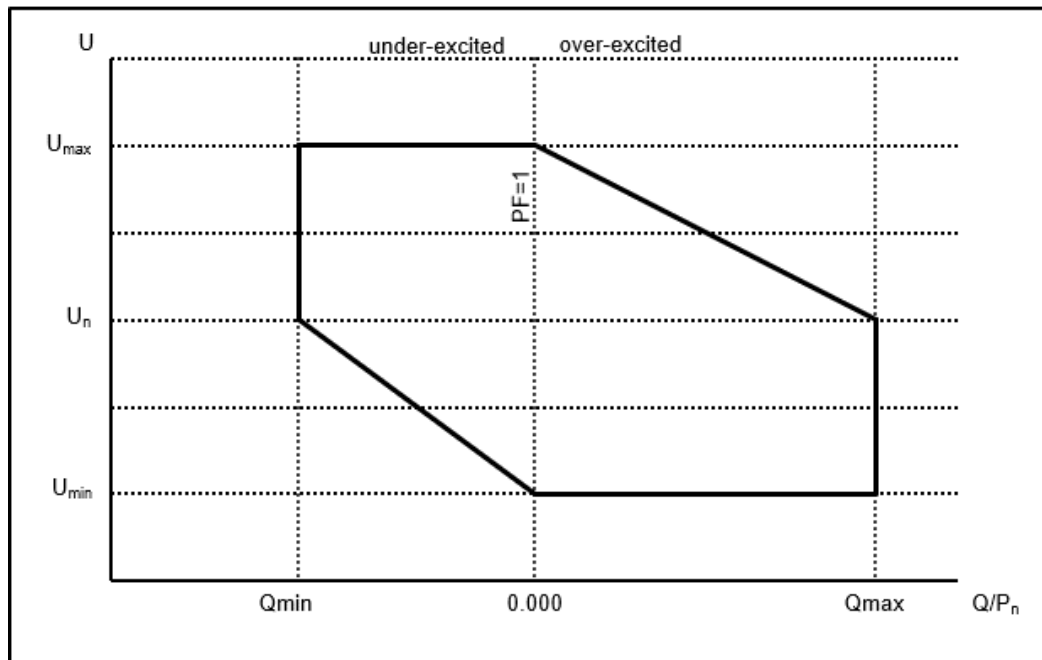


Figure 9: Voltage dependence of reactive power capability for BESF of category B and C at the POC

8.3. Reactive power and voltage control

8.3.1. General

- (1) The *BESF* of category *B* and *C* shall be equipped with reactive power control functions capable of controlling the reactive power at the *POC* contributing to control reactive power supplied by the *BESF* at the *POC* as well as a voltage control function capable of controlling the voltage at the *POC* via orders using setpoints and gradients for the specified parameters.
- (2) The three reactive power control functions, *Q* control, power factor control and voltage control are mutually exclusive, which means that only one of the three functions mentioned below can be activated at a time.
- (3) The reactive power capabilities are specified at the *POC* unless otherwise specified by the *SO*
- (4) The control function and applied parameter settings for reactive power and voltage control functions shall be determined by the *NSP* in collaboration with the *SO* and implemented by the *BESF* owner. The agreed control functions and parameters shall be documented in the *operating agreement*.
- (5) Unless otherwise agreed with the *SO* or *NSP*, *BESF* of categories *A3*, *B* and *C* shall be equipped with remote control facilities according to Table 6 below.

Table 6: Automatic reactive power and voltage control functions for all TS or DS-connected *BESF* of category *A3*, *B* and *C*

Function	A1	A2	A3	B1	B2	C
Power factor control	-	-	✓	✓	✓	✓
Reactive power control	-	-	-	✓	✓	✓
Voltage control	-	-	-	✓	✓	✓
Remote control capability	-	-	✓	✓	✓	✓

8.3.2. Power factor (PF) control

- (1) *BESF* shall receive the power factor setpoint from the *SO* or *NSP*. The setpoint determines the required power factor at the *POC*.
- (2) Category *A3* *Power Plants* shall be operated at unity power factor at the *POC*, unless otherwise specified by the *NSP*.
- (3) Following a small disturbance, the *BESF* shall change its reactive power output so that the actual power factor equals the power factor setpoint within $\pm 2\%$. The accuracy shall be measured as a deviation of the actual reactive power from the required reactive power (at the required power factor), as a percentage of the maximum required reactive power.
- (4) The response time shall be adjustable. Unless otherwise specified by the *SO* and agreed with the *BESF* owner, *NSP* or their agent, the settling time, following a small disturbance, shall be in the range 10-30s.

8.3.3. Reactive power (Q) control

- (1) *BESF of category B and C* shall receive the reactive power setpoint from the *SO* or *NSP*. The setpoint determines the required reactive power at the POC.
- (2) Following a small disturbance, the *BESF* shall change its reactive power to equal the reactive power setpoint within $\pm 2\%$. The accuracy shall be measured as a deviation of the actual reactive power from the required reactive power, as a percentage of the maximum required reactive power.
- (3) The response time shall be adjustable. Unless otherwise specified by the *SO* and agreed with the *BESF* owner, *NSP* or their agent, the settling time, following a small disturbance, shall be in the range 10-30s.

8.3.4. Voltage (V) control

- (1) The *BESF of category B and C* shall receive the voltage setpoint and the *voltage droop* setpoint from the *SO* or *NSP*.
- (2) *BESF of category B and C* shall be capable of automatic voltage control according to a *droop* function, as shown in Figure 10, within its reactive power capability range. In this context, *droop* is the voltage change (p.u.) caused by a change in reactive power (p.u.).
- (3) Following a small disturbance, the *BESF* shall change its reactive power so that the voltage equals the required voltage, as per the voltage setpoint and droop setpoint, within $\pm 0.2\%$. The accuracy shall be measured as a deviation between the actual voltage and the required voltage, as a percentage of the rated voltage. Unless otherwise agreed with the *SO* or *NSP*, the settling time shall be in the range 10 to 30s.

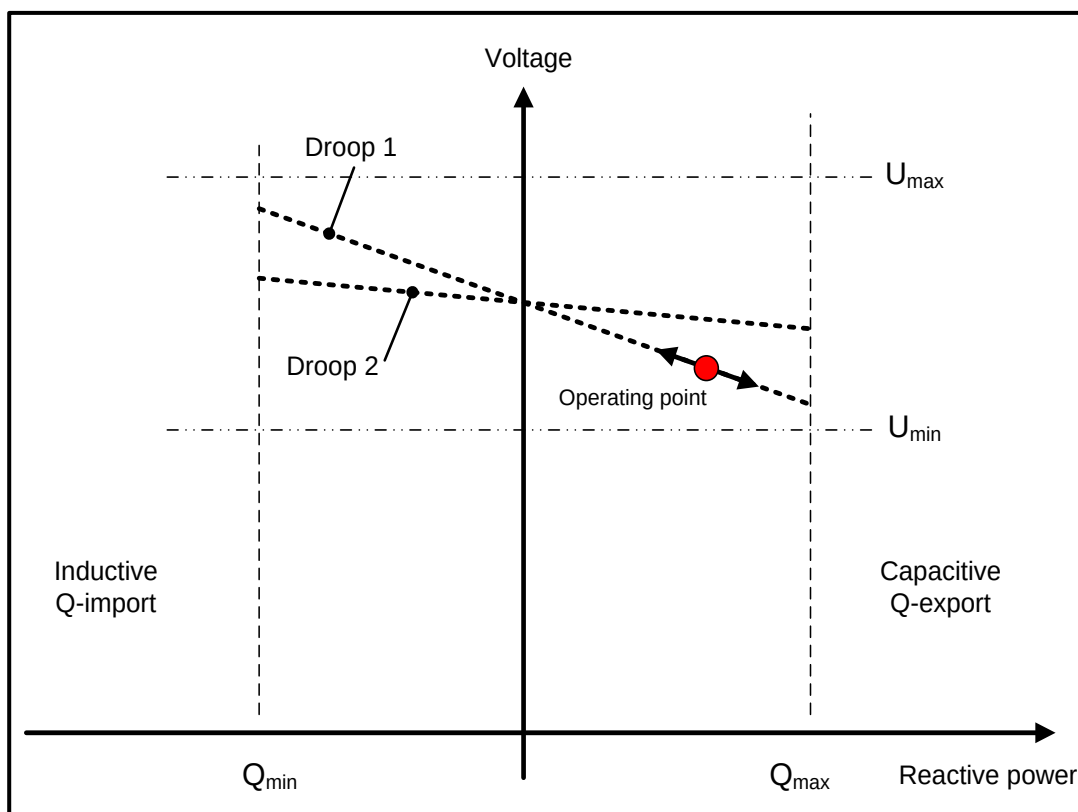


Figure 10: Voltage control for BESF of Category B and C at the POC

9. Power Quality

9.1. Category A and B1 BESF

- (1) For BESF of categories A and B1, the power converters' harmonic current emissions and their total harmonic distortion (measured in accordance with IEC standards) shall not exceed the limits shown in Table 7.
- (2) For BESF categories A and B1, the voltage fluctuation and flicker shall not exceed the limits specified in SANS61000-3-3 (for equipment rated less than or equal to 16 A per phase) and in SANS61000-3-5 (for equipment rated greater than 16 A per phase).

Table 7: Harmonic current limits for power converters of category A and B1 BESF

Harmonic order	$3 \leq h < 9$	$11 \leq h < 15$	$17 \leq h < 21$	$23 \leq h < 33$	$33 \leq h \leq 50$	Even	THD
Percentage of rated current (odd harmonics)	<4.0	<2.0	<1.5	<0.6	<0.3	25% of odd harmonics	<5.0

9.2. Category B2 and C BESF

- (1) The *BESF* of category B2 and C shall not cause voltage and current emissions at the *POC* to exceed the apportioned emission limits specified by the *NSP*.
- (2) The *NSP* shall determine the apportioned emission levels based a fair and transparent process, in accordance with international and local standards.
- (3) For *BESF* of categories B2 and C, current harmonic emissions at individual harmonic orders may exceed the apportioned limits by up to 50%, provided that the group harmonic distortion levels are within the limits shown in Table 8, where *h* is the harmonic order.
- (4) For grid code compliance the *BESF* shall monitor and report on power quality using an IEC 61000-4-30 Class A power quality monitoring device. The reporting will be done to prove compliance at the *POC* against the *NSP* requirements specified in the system supply agreements. The power quality parameters to be reported on must at least include:
 - flicker
 - harmonics
 - unbalanced voltages
- (5) Appendix 6 provides guidance to *BESF* owners on a method to prove compliance to the *Power Quality* requirements as set out by the *NSP*.
- (6) The *BESF* owner shall ensure that the *BESF* is designed, configured and implemented in such a way that the specified emission limit values are not exceeded.
- (7) The *BESF* shall not cause the impedance at the *POC* to exceed the limit $Z(h)$, as defined below. This limit applies for all frequencies up to the 50th harmonic and for all typical network operating conditions.

$$Z(h) = 3 \cdot h \cdot \frac{V^2}{S}$$

where *Z* is the harmonic impedance, *h* is the harmonic number, *V* is the nominal voltage (line-to-line) in kV and *S* is the fault level in MVA.

- (8) To assist with the maximum resonance of 3 times as per clause (7) above, no *BESF* may connect equipment, e.g. shunt capacitor banks, that will cause a resonance of more than 3 times at the *POC* at any frequency.
- (9) The *NSP* shall manage the consequences of the network impedance at the *POC* exceeding the limit in clause (7), where such exceedances exist prior to the first commissioning of the *BESF* or occur following the development of the *TS* or *DS* after the first commissioning of the *BESF*.

- (10) The Transmission and Distribution network service providers shall use reasonable endeavours to furnish the *BESF* with a reliable and continuous connection for the delivery of electrical energy up to the *POC*. The network operators do not guarantee that the continuity and voltage quality of the connection will always be maintained under all contingencies. It is therefore incumbent upon the *BESF* to take adequate measures to protect the *BESF* against any losses and/or damage arising from frequency deviations, connection/supply interruptions, voltage variations (including voltage dips), voltage harmonics, voltage flicker, voltage unbalance, voltage swells and transients, undervoltage and overvoltage's in the connection. It is also mandatory upon the *BESF* to take such necessary measures so as not to cause any damage to the *TS* and *DS*.

Table 8: Harmonic distortion limits for *BESF* of categories B2 and C

Harmonic distortion band	Harmonic order band	Group harmonic current distortion limit
A	$2 \leq h \leq 13$	$HD_{IlimitA} = \sqrt{\sum_{h=2}^{13} I_{limit}(h)^2}$
B	$14 \leq h < 25$	$HD_{IlimitB} = \sqrt{\sum_{h=14}^{25} I_{limit}(h)^2}$
C	$26 \leq h \leq 39$	$HD_{IlimitC} = \sqrt{\sum_{h=26}^{39} I_{limit}(h)^2}$
D	$40 \leq h \leq 50$	$HD_{IlimitD} = \sqrt{\sum_{h=40}^{50} I_{limit}(h)^2}$

10. Protection and Fault levels

- (1) This section shall apply to *BESF* of all categories.
- (2) Internal Protection functions shall be available to protect the *BESF* as well as the *TS* and *DS* and to ensure a stable *TS* and *DS*.
- (3) Protection functions for internal *BESF* faults shall not jeopardize *VRT* capabilities
- (4) The *BESF owner* shall ensure that a *BESF* is dimensioned and equipped with the necessary protection functions so that the *BESF* is protected against damage due to internal faults and incidents in the *TS* and *DS*.
- (5) The *BESF* of all categories shall be equipped with an effective *LOM* detection, functional in all system configuration and operation modes and capability to shut down generation of power in such condition within 2 seconds. Unintended *BESF* islanded operation with part of the *TS* or *DS* is not permitted unless specifically agreed with the *NSP*. Vector shift relay as detection scheme are prohibited.

- (6) The *NSP* or the *SO* may request that the set values for protection functions be changed following commissioning if it is deemed to be of importance to the operation of the *TS* and *DS*. However, such change shall not result in the *BESF* being exposed to negative impacts from the *TS* and *DS* lying outside of the design requirements.
- (7) The *NSP* shall inform the *BESF* owner of the highest and lowest short-circuit current that can be expected at the *POC* as well as any other information about the *TS* and *DS* as may be necessary to co-ordinate the *BESF*'s protection functions.

11. BESF Availability, and Supervisory Control and Data Acquisition

11.1. General

- (1) This section shall apply to *BESF* of all categories.
- (2) The signal list is a list of the signals/information that must be exchanged between the parties that control and monitor a *BESF* as per Figure 11. The signals specified must be present at the *PCOM* interface
- (3) The signals indicated in clauses below are the minimum requirement that the *BESF owner* must provide per category; additional site specific signals may also be accommodated in agreement with the relevant *SO/NSP*.
- (4) All digital signals reported to the *Gateway* shall be time-stamped (UTC +2:00) to an accuracy of +/- 1 millisecond and shall be reported within 1 second of any change.
- (5) All control signals sent from *SO/NSP* to the *BESF* must have an acknowledgement returned back to the *SO/NSP* as per applicable protocol in section 12.
- (6) Timestamps on analogue changes are not required.
- (7) All analogue input changes shall be reported to the *Gateway* within 1 second of any change greater than or equal to the following:
 - a) Frequency shall be updated when the value changes by 0.01 Hz or more.
 - b) Power factor values shall be updated when the value changes by 0.01 or more.
 - c) Active power values shall be updated when the value changes by 1 % or more of rated power. If rated power is 50 MW or more then values shall be updated when the value changes by 0.5 MW or more.
 - d) Actual Ramp Rate shall be updated when the value changes by 1 MW/min or more.
 - e) All other analogues shall be updated when the value changes by 1% or more of the full-scale or nominal value.

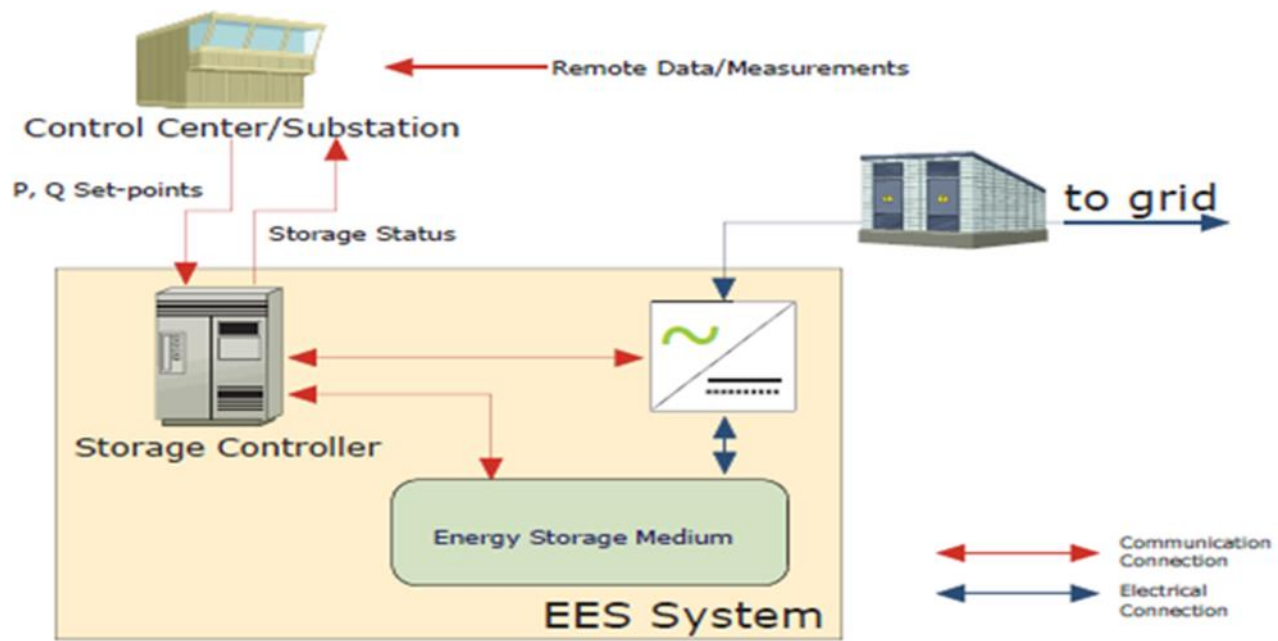


Figure 11 - EES System model with interfaces

11.2. Signals from the BESF to be made available at the POC/PCOM

item	Description	Criteria	Category									
	Cat B1 & A3 for own use BESF plant (the POC is not at a utility substation) and A3 (LV) is included only for the signal list		C	B2	B1	B1 MV	B1 MV own use	A3 MV	A3 MV own use	A3 LV	A2	A1
1	Actual Active Power Export/Produce (+) Import/Absorb (-) at POC	MW or KW	X	X	X	X	X	X	X			
2	Actual ramp rate of entire BESF Power Gradient Constraint ramp rate (7.2)	MW/min	X	X	X	X		X				
3	Actual Reactive Power Export/Produce (+) Import/Absorb (-) at POC	-/+Mvar	X	X	X	X		X				
4	Actual power current export / import (A)	Amps / phase	X	X	X	X		X				
5	Voltage measured at point of connection	Volts / phase	X	X	X	X		X				
6	BESF POC breaker status (a per NSP specifications)	open / closed	X	X	X	X		X				
7	BES plant in shutdown mode	yes/no	X	X	X	X		X				
8	frequency at POC	Hz	X	X								
9	Freq response mode status for freq support	on/off	X	X	X	X						
10	Power factor - measured in point of connection	cos (phi)	X	X	X	X		X				
11	Curtailment mode status - Absolute Power Control	on/off	X	X	X	X						
12	Curtailment setpoint = Absolute Power Control limiter	MW	X	X	X	X						
13	Active Power gradientt constraint mode (P)	on/off	X	X	X	X						
14	Active Power gradient Up Ramp rate setpoint feedback	MW/min	X	X	X	X						
15	Active Power gradient Down Ramp rate setpoint feedback	MW/min	X	X	X	X						
16	Reactive power control mode (Q)	on/off	X	X	X	X		X				
17	Reactive power control setpoint feedback	Mvar	X	X	X	X		X				
18	Reactive power lower limit	Mvar	X	X	X	X		X				
19	Reactive power upper limit	Mvar	X	X	X	X		X				
20	Power Factor control mode (PF)	on/off	X	X	X	X		X				
21	Power factor control setpoint	cos (phi)	X	X	X	X		X				
22	voltage control mode (V)	on/off	X	X	X	X						
23	Voltage control setpoint -raise or lower value	Volts	X	X	X	X						
24	voltage control setpoint feedback	Volts	X	X	X	X						

11.3. Control Signals/Commands send to the BESF

item	Description	Criteria	Category									
	Cat B1 & A3 for own use BESF plant (the POC is not at a utility substation) and A3 (LV) is included only for the signal list		C	B2	B1	B1 MV	B1 MV own use	A3 MV	A3 MV own use	A3 LV	A2	A1
1	Stop command (per NSP specifications).	on/off					X	X	X	X	X	X
2	Hold command (per NSP specifications)	on/off					X	X	X	X	X	X
3	BESF POC breaker trip command (per NSP specifications)	on/off	X	X	X	X		X				
4	All commands/setpoints as per signal list item 9-24											

12. Communications Specifications

12.1. Communication Interface Gateway

- (1) *The communication interface Gateway shall be at the PCOM and shall be owned, operated and maintained by the BESF owner.*
- (2) *The communication interface Gateway shall be compatible with the SCADA Master Station protocols as per section 12.2.1 and shall be tested as per the process described in Appendix 3.*
- (3) *This interface provides to the ESS system input signals and receive from it output signals, both in a form directly usable for control or measurement purposes, or which enables digital communication with other systems or devices. The communication interface is usually designed to a specific standard and used for transmitting control and measurement data.*
- (4) *The Gateway shall have an availability of 99.99%*

12.2. Protocols for Information Exchange

- (1) *Only standardized protocols as specified below in section 12.2.1 shall be used for SCADA information exchange between the BESF and the Gateway and the NSP or SO.*
- (2) *The BESF control system shall be capable of reliably exchanging system status and data with the NSP or SO via the Gateway. The BESF owner is responsible for the communication infrastructure to connect the required number of Gateway ports to the NSP or TNSP telecommunications infrastructure in the NSP or TNSP substation.*

12.2.1. SCADA Protocol between the Gateway and the NSP or SO

- (1) *Standardized communication should be used for the communication between EES system and stakeholders. Common standards used for grid management involving control centers are IEC 351-56-14, IEC 60870-5-101/104, IEC 60870-6, IEEE 1815 (DNP3) or IEC 61850. (IEV 351-56-14)*
- (2) *For BESF which is not connected directly to a Utility s/s the communication interface and protocol shall be other international standards.*

12.2.2. Protocols applied within the BESF system

- (1) *The BESF owner may select proprietary protocols for communication within the BESF plant control system if compliance to the cyber security requirements specified by the SO can be verified.*

- (2) For *BESF* of category *B* & *C*, the system shall use GPS based time stamping for all digital signals at the Intelligent Electronic Device (IED) or plant controller level (if IED is not applicable) which will be propagated to the *SCADA master station* with date and time to 1 millisecond accuracy in the defined protocol as per section 12.2.1.

12.3. Telecommunication Interface Requirements

- (1) The telecommunication interface to the *NSP* or *SO* may vary from substation to substation of the *NSP*. The *BESF owner* shall provide the equipment to interface with the *NSP* telecommunication infrastructure.
- (2) If not otherwise specified the default telecommunication interface at the Gateway shall be via single-mode fiber optic (FO) cable. Other telecommunication interfaces if necessary shall be agreed upon by the parties.
- (3) The communication link from the *BESF* Gateway and up to the *NSP's* telecommunication interface shall be owned and maintained by the *BESF owner*.
- (4) To protect the communications bandwidth especially on RF Radio channels/Cellular Networks the values used by the Gateway to report analogue changes to the *NSP* may be set higher than the values listed in section 11 (8).
- (5) For *BESF* which is not connected directly to a Utility substation the communication interface and protocol shall be to other international standards.

12.4. Operational Communications

- (1) Each *BESF owner* of *Category B&C* shall be responsible for providing the following to the *NSP* or *SO* for operational purposes:
 - (a) one mobile telephone contact number and
 - (b) One email address that shall be continuously attended to and responded by authorised personnel within a reasonable time frame (5 minutes).
- (2) The *SO/NSP* shall use a voice recorder for historical recording of all operational voice communication with the *BESF owner*. These records shall be available for at least three (3) months. The *SO/NSP* shall make the voice records of an identified incident in dispute available within a reasonable time frame after a request from the *BESF owner* and/or *NERSA*.

12.5. Data Communications Requirements

- (1) The necessary communications links, communications protocol and the requirement for analogue or digital signals shall be specified by the *NSP* as appropriate before a connection agreement is signed with the *BESF owner*.
- (2) Where signals or indications required to be provided by the *BESF owner* become unavailable or do not comply with applicable standards due to failure of the *BESF* equipment or any other reason under the control of the *BESF* operator, the *BESF owner* shall restore or correct the signals and/or indications within a reasonable time frame from the request from the *NSP/SO*.

13. Testing and Compliance Monitoring

13.1. General

- (1) All *BESF owners* shall demonstrate full compliance to all applicable requirements specified in this grid connection before being allowed to connect to the TS or DS and operate commercially and as specified in any part of the *Grid Code*.
- (2) *BESF owners* shall demonstrate compliance through the provision of design information, study reports, test reports and as-built documentation.
- (3) For *BESF* of category *A1* and *A2* the owner and manufacturer's declaration of compliance as well as independent testing can be used to demonstrate compliance.
- (4) The *BESF owners* shall keep records relating to the compliance of the *BESF* with each section of this grid connection code, or any other *code* applicable to that *BESF*, setting out such information that the *SO* or responsible *NSP* reasonably requires for assessing power system performance, including actual *BESF* performance during abnormal conditions. Records shall be kept for a minimum of 5 years (unless otherwise specified in the *code*) commencing from the date the information was created.
- (5) The *BESF owners* shall continuously monitor its compliance with all the connection conditions of this code.
- (6) Following a modification to a *BESF* in accordance with section 14, the *BESF owners* shall execute the studies and tests required to confirm compliance with this code.

13.2. Studies and tests

- (1) *BESF owners* shall perform simulation studies and tests, including those listed in Table 12, to demonstrate compliance with this Code.
- (2) Simulation studies shall be based on validated models of the *BESF*. *BESF owners* shall provide evidence of the validity of the simulation models.
- (3) Each *BESF owner* of *Category A3* and above shall submit to the *SO* or *NSP* a detailed test procedure, emphasizing system impact, for each relevant part of this code prior to every test.
- (4) The continuous monitoring of compliance shall be in line with the procedures of the “routine” studies and tests described in Table 12.
- (5) The scope of the studies and tests that are required following a plant modification shall depend on which requirements are affected by the modification and shall be agreed with the *SO* or relevant *NSP*.
- (6) The *SO* or responsible *NSP* may provide additional guidelines about the execution of simulation studies and tests.

Table 12: Minimum set of studies and tests per *BESF* category

Study or test	BESF Category
<u>Active power capability and associated control</u>	
Simulation study showing frequency response capability and performance	Where function is required
Test demonstrating frequency response capability and performance	Where function is required
Test demonstrating absolute power constraint function	Where function is required
Test demonstrating delta power constraint function	Where function is required
Test demonstrating power gradient function	Where function is required
<u>Reactive power capability and associated control (Q, P_f and V)</u>	
Simulation study showing reactive power capability and performance	A3, B and C,
Test showing control capability and performance	A3, B and C,
<u>Tolerance of Voltage Deviations</u>	
Simulation study showing voltage response capability and performance	A3, B and C,
<u>Power quality</u>	
Simulation study showing power quality performance	B2 and C,
Tests showing power quality performance	B2 and C,
<u>Protection</u>	
Study on protection functions and settings	A, B and C,

13.3. Non-compliance determined by the *BESF Owner*

- (1) If a *BESF owner* determines, from tests or otherwise, that its *BESF* is not complying in any respect with one or more sections of this code, then the *BESF owner* shall:
 - (a) notify the *SO* or *NSP* of that fact within one hour after becoming aware of the non-compliance,
 - (b) promptly advise the *SO* or *NSP* of the remedial steps it proposes to take to ensure that the relevant *BESF* (as applicable) can comply with this code and the proposed timetable for implementing those steps
 - (c) diligently take such remedial action as will ensure that the relevant *BESF* (as applicable) can comply with this code; the *BESF owner* shall regularly report in writing to the *SO* or *NSP* on its progress in implementing the remedial action, and
 - (d) after taking remedial action as described above, demonstrate to the satisfaction of the *SO* or *NSP* that the relevant *BESF* (as applicable) is then complying with this code.

13.4. Non-compliance suspected by the System Operator

- (1) If at any time the *SO* or responsible *NSP* suspects that a *BESF* is not complying with a requirement in this code, then the *SO* or *NSP* shall notify the relevant *BESF owner* of such non-compliance by issuing a *non-conformance report* (as referred to in the *Governance Code*), specifying the requirement concerned and the basis for the *SO*'s or *NSP*'s suspicion.
- (2) The *SO* or responsible *NSP* shall specify the remedial action required from the *BESF owner* as well as the time frames within which to comply with this code.
- (3) Any dispute arising out of such a *non-conformance report* shall be resolved in terms of the dispute resolution procedure described in the *Governance Code*.
- (6) The *SO* or responsible *NSP* may issue an instruction requiring a *BESF owner* to carry out a test to demonstrate that the relevant *BESF* complies with the *Grid Code* requirements. A *BESF owner* may not refuse such an instruction, provided it is issued timeously and there are reasonable grounds for suspecting non-compliance.

14. Modifications

- (1) If a *BESF owner* proposes to change or modify the *BESF* in a manner that could reasonably be expected to either affect its ability to comply with this code, or changes the performance, information supplied, settings, etc., then that *BESF owner* shall follow the following process:
 - (a) submit a proposal notice to the *SO* or relevant *NSP* which shall

- contain detailed plans of the proposed change or modification
 - state when the *BESF owner* intends to make the proposed change or modification, and
 - set out the proposed tests to confirm that the relevant *BESF* as changed or modified to operate in the manner contemplated in the proposal, can comply with this code.
- (b) If the *SO* or relevant *NSP* disagrees on reasonable grounds with the proposal submitted, it shall provide the relevant *BESF owner* with reasons, and the *SO* or *NSP* and the relevant *BESF owner* shall promptly meet and discuss the matter in good faith in an endeavour to resolve the disagreement.
- (c) The *BESF owner* shall ensure that an agreed change or modification to a *BESF* or to a subsystem of a *BESF* is implemented in accordance with the relevant proposal agreed to by the *SO* or *NSP*.
- (d) The *BESF owner* shall notify the *SO* or *NSP* promptly after an agreed change or modification to a *unit* has been implemented.
- (e) The *BESF owner* shall confirm that a change or modification to the *BESF*, as described above conforms to the relevant proposal by conducting the relevant tests, in relation to the connection conditions, promptly after the proposal has been implemented.
- (f) A *BESF owner* shall provide the *SO* or *NSP* with a report in relation to any compliance test (including test results of that test, where appropriate), within 20 business days after such test has been conducted.

15. Provision of Data and Electrical Dynamic Simulation Models

- (1) Exchange of the relevant information between the parties involved in the grid connection application process is one of the fundamental requirements for a successfully process.
- (2) The following information must as a minimum be exchanged for *BESF* category B2 and C.

15.1. Data provision

- (1) *BESF* information exchange is a time-based process:
- (a) Before grid connection date (GCD) (6 – 12 months)
- (i) The following information shall be submitted by the *BESF owner* to the *SO* and *NSP* or *TNSP*, as applicable:
- Physical location of the *BESF* (including the GPS coordinates)
 - Site Plan

- MW output per *inverter* (P_{nd} and P_{na});
 - Initial phase MW value (P_{nd} and P_{na});
 - Final phase MW value and timelines (P_{nd} and P_{na});
 - Any other information that the *NSP* may reasonably require
- (ii) For *BESF* category B2 and C a generic *BESF* owner dynamic simulation model from accepted industrial standards including a generic *BESF* facility controller or *BESF* unit model from the original equipment manufacturer (OEM) and selected *BESF* technology data. for dynamic modelling purpose as per type approval tests result;
- (iii) For the detailed *BESF* design, the *NSP* shall make available to the *BESF* owner or its agent at least the following information:
- *Point of Connection* and the *Point of Common Coupling* including the nominal voltages,
 - Expected fault levels
 - The *network service provider's* connection between the *Point of connection* and the *BESF*,
 - The busbar layout of the *PCC* and *POC* substations,
 - The portion of the network service provider's grid that will allow accurate and sufficient studies to design the *BESF* to meet the Grid Code. This information shall include:
 - Positive and zero sequence parameters of the relevant network service provider's transmission and distribution, transformers, reactors, capacitors and other relevant equipment,
 - The connection of the various lines transformers, reactors and capacitors etc.
- (b) Before Commercial operation date (COD) (after facility completion and including commissioning)
- (i) During this stage, the *BESF owner* of category B2 and C shall provide information on:
- Grid code compliance study report which include reactive power capability, fault ride through and harmonic studies
 - Final design OEM unit model and dynamic modelling data per the OEM type approval tests result

(c) After commercial operation of the BESF (within 6 months after COD)

(i) During this stage, the *BESF owner* of category B2 and C shall provide the following information :

- A validated *BESF* electrical dynamic simulation model using commissioning test data and measurements, including generating and charging modes
- This final validated model shall be comprised of the following; generic or OEM
- *BESF controller* and OEM unit models
- *BESF* model validation report
- *BESF controller* manual if not incorporated in the validation report
- Site test measurement data in the format agreed between the *BESF owner* and the *NSP*, *NTC* or *SO*, as applicable.

(1) In addition, the *BESF Owner* shall provide the *SO* or *NSP* with data as prescribed in Appendix 1.

15.2. Electrical simulation model

- (1) The *SO* and *NSPs* require suitable and accurate dynamic models, in the template specified by the requesting party applying for a connection to the *DS* or *TS*, in order to assess reliably the impact of the *BESF* proposed installation on the dynamic performance and security and stability of the power system.
- (2) The required dynamic models must operate under *RMS* and *EMT* simulation to replicate the performance of the *BESF* for analysis of the following network aspects:
- (a) *BESF* impact on network voltage stability
 - (b) *BESF* impact on *QOS* at *POC*
 - (c) *BESF* switching transients impact on network performance
 - (d) *BESF* impact on breakers *TRV* (Transient Recovery Voltage)
 - (e) *BESF* impact on network insulation co-ordination requirements
 - (f) *BESF* impact on network protection co-ordination
 - (g) *BESF FRT* (Fault Ride Through) capability for different types of faults and positions
 - (h) *BESF* response to various system phenomena such as:

- (i) switching on the network
 - (ii) power swings
 - (iii) small signal instabilities
- (3) Generic instead of type tested *EMT* models can be accepted on condition that they represent *BESF* performance with frequency spectrum from 0 to 1kHz with an accuracy level as specified in Appendix 7.
- (4) *EMT* models must include all parameters required for *EMT* simulations such as positive, negative and zero sequence impedances for all elements, magnetising curves, losses and tap changer data for transformers as well as positions of surge arresters and their V/I characteristics. *EMT* models must also include all protection and control functions of the plant.
- (5) The dynamic modelling data shall be provided in a format as may be agreed between the *BESF* owner and the *NSP* or *SO*, as applicable.

16. Reporting to NERSA

- (1) The *BESF* owner of category *B2* and *C* installations shall design the system and maintain records so that the following information can be provided to the *NERSA* on a monthly basis in an electronic spread sheet format:
- (a) Non-renewable/supplementary fuel used by the power plant as outlined under Supplementary Fuel Specification schedule of the *PPA* during the month.
 - (b) Day ahead forecast of hourly output/input energy to the grid and hourly availability.
 - (c) Actual hourly availability and output/input energy to the grid that occurred
 - (d) Actual hourly reactive energy output/input to the grid that occurred
 - (e) The log of control set point and mode changes.
 - (f) Actual hourly auxiliary electricity imports from all sources if applicable.
 - (g) Direct monthly emissions per unit of electricity generated by the *BESF* (tCO₂/kWh).
 - (h) Any curtailed energy during the month.
 - (i) If operated in tandem with a renewable generator or is situated behind the *POC* of a *renewable generator* as defined in the grid connection code for renewable power plant code then reporting must be separate for both units and in addition there must be a combined report for the combined operation impact on grid at *POC*.

- (2) These reports are to be submitted before the 15th of the following month to IPSreports@nersa.org.za
- (3) These reports should also include details of incidents relating any unavailability of the network which prevented the *BESF* from generating and any incidents where their right to self-dispatch was impinged upon where the *PPA* gives them a right to self-dispatch.

Appendices

Appendix 1 Documentation

A1.1 Master Data

Description	Text
Identification:	
Name of <i>electricity supply undertaking</i>	
Plant name	
ID number	
Planned commissioning	
Technical data:	
Manufacturer	
Type designation (model)	
Type approval	
Approval authority	
Installed kW (<i>rated power</i>)	
Cos φ (<i>rated power</i>)	
Cos φ (<i>20% rated power</i>)	
Cos φ (<i>no load</i>)	
3-phase short-circuit current immediately in front of the <i>power plant</i> (RMS)	
<i>Point of connection</i>	
Voltage level	

A1.2 Technical Documentation

A1.2.1 Step-Up Transformer

Description	Value
Make	
Type	
Comments	

Description	Symbol	Unit	Value
Nominal apparent power (1 p.u.)	S_n	MVA	
Nominal primary voltage (1 p.u.)	U_p	kV	
Nominal secondary voltage	U_s	kV	
Coupling designation, eg Dyn11	-	-	
Step switch location	-	-	Primary side ☐ Secondary side ☐
Step switch, additional voltage per step	du_{tp}	%/trin	
Step switch, phase angle of additional voltage per step:	ϕ_{tp}	degree/st ep	
Step switch, lowest position	n_{tpmin}	-	
Step switch, highest position	n_{tpmax}	-	
Step switch, neutral position	n_{tp0}	-	
Short-circuit voltage, synchronous	u_k	%	
Copper loss	P_{cu}	kW	
Short-circuit voltage, zero system	u_{k0}	%	
Resistive short-circuit voltage, zero-sequence system	u_{kr0}	%	
No-load current	I_0	%	
No-load loss	P_0	%	
Internal resistance of discharge	R_{dis}	Ω	
Charging Capacity	Q_{bat}	kWh	
Maximum Charging Capacity	$Q_{bat (max)}$	kWh	
Minimum Charging Capacity	$Q_{bat (min)}$	kWh	
Efficiency of charge.	η_{ch}	%	
Efficiency of discharge	η_{dis}	%	

A1.2.2 Single Line Diagram Representation

(1) This applies to all *BESF* of category A3, B and C. The *SO*, *NSP* or local *network operator* may request that a single-line diagram representation be provided for *BESF* of category A1 and A2.

(2) A single-line diagram representation of the plant shall be created, with indication of *POC*, metering points, including settlement metering, limits of ownership and operational supervisor limits/limits of liability. In addition, the type designation for the switchgear used shall be stated so as to make it possible to identify the correct connection terminals.

(3) In instances when a single-line diagram representation is included in the grid use agreement between *BESF Owner* and *SO*, the *grid connection agreement* can be enclosed as documentation.

A1.2.3 PQ Diagram

(1) This applies to all *BESF* of category A3 B and C. The *SO*, *NSP* or local *network operator* may also request that a PQ diagram representation be provided for *BESF* of category A1 and A2.

Appendix 2 Compliance test specifications

A2.1 Introduction

This section specifies the procedures to be followed in carrying out testing to verify compliance with this *Code*.

A2.2 Test procedures

A2.2.1 - BESF protection function verification		
Parameter	Reference	Description
Protection function and settings	Section 10	APPLICABILITY AND FREQUENCY All new <i>BESF</i> coming on line or at which major refurbishment or upgrades of protection systems have taken place. Routine review: Confirm compliance every 6 years. PURPOSE To ensure that the relevant protection functions in the <i>BESF</i> are coordinated and aligned with the system requirements. PROCEDURE 1. Establish the system protection function and associated trip level requirements from the <i>SO</i> or relevant <i>NSP</i>. 2. Derive protection functions and settings that match the <i>BESF</i> and system requirements. 3. Confirm the stability of each protection function for all relevant system conditions. 4. Document the details of the trip levels and stability calculations for each protection function. 5. Convert protection tripping levels for each protection function into a per <i>unit</i> base. 6. Consolidate all settings in a per <i>unit</i> base for all protection functions in one document. 7. Derive actual relay dial setting details and document the relay setting sheet for all protection functions. 8. Document the position of each protection function on one single line diagram of the generating <i>unit</i> and associated connections. 9. Document the tripping functions for each tripping function on one tripping logic diagram. 10. Consolidate detail setting calculations, per unit setting sheets, relay setting sheets, plant base information on which the settings are based, tripping logic diagram, protection function single line diagram and relevant protection relay manufacturers' information into one document. 11. Submit to the <i>SO</i> or relevant <i>NSP</i> for its acceptance and update. Review: 1. Review Items 1 to 10 above. 2. Submit to the <i>SO</i> or relevant <i>NSP</i> for its acceptance and update. 3. Provide the <i>SO</i> or relevant <i>NSP</i> with one original master copy and one working copy. ACCEPTANCE CRITERIA

		All protection functions are set to meet the necessary protection requirements of the <i>BESF</i> with a minimal margin, optimal fault clearing times and maximum plant availability. Submit a report to the <i>SO</i> or relevant <i>NSP</i> one month after commissioning and six-yearly for routine tests.
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A.2.2.2 - BESF protection integrity verification		
Parameter	Reference	Description
Protection integrity	Section 10	APPLICABILITY AND FREQUENCY All new <i>BESF</i> coming on line or after major works of refurbishment of protection or related plant. Also, when modification or work has been done to the protection, items 2 to 5 must be carried out. This may, however, be limited to the areas worked on or modified. Routine review: All <i>BESFs</i> on: item 1 below: Review and confirm every 6 years Item 2, and 3 below: at least every 12 years. PURPOSE To confirm that the protection has been wired and functions according to the specifications. PROCEDURE 1. Apply final settings as per agreed documentation to all protection functions. 2. With the <i>unit</i> off load and de-energized, inject appropriate signals into every protection function and confirm correct operation and correct calibration. Document all protection function operations. 3. Carry out trip testing of all protection functions, from origin (e.g. Buchholz relay) to all tripping output devices (e.g. HV breaker). Document all trip test responses. 4. Apply short-circuits at all relevant protection zones and with <i>BESF</i> at nominal speed excite <i>generator</i> slowly, record currents at all relevant protection functions and confirm correct operation of all relevant protection functions. Document all readings and responses. Remove all short-circuits. 5. With the <i>BESF</i> at nominal production. Confirm correct operation and correct calibration of all protection functions. Document all readings and responses. Review: Submit to the <i>SO</i> or relevant <i>NSP</i> for its acceptance and update. ACCEPTANCE CRITERIA All protection functions are fully operational and operate to required levels within the relay <i>OEM</i> allowable tolerances. Measuring instrumentation used shall be sufficiently accurate and calibrated to a traceable standard. Submit a report to the <i>SO</i> or relevant <i>NSP</i> one month after test.

A2.2.3 - BESF active power control capability verification		
Parameter	Reference	Description
Active power control function and operational range	Section 6 & 7 depending on category	APPLICABILITY All new <i>BESFs</i> coming on line and after major modifications or refurbishment of related plant components or functionality. Routine test/reviews: Confirm compliance every 6 years. PURPOSE To confirm that the active power control capability specified is met. PROCEDURE The following tests shall be performed within an active power level range of at least 0.2p.u.or higher 1. The <i>BESF</i> will be required to regulate the active power to a set of specific setpoints within the design margins. 2. The <i>BESF</i> will be required to obtain a set of active power setpoints within the design margins with minimum two different gradients for ramping up and two different gradients for ramping down. 3. The <i>BESF</i> will be required to maintain as a minimum two different set levels of spinning reserve within the design margins. 4. The <i>BESF</i> will be required to operate as a minimum to limit active power output according to two different absolute power constraint set levels within the design margins. 5. The <i>BESF</i> will be required to verify operation according to as a minimum two different parameter sets for a frequency response curve within the design margins. ACCEPTANCE CRITERIA 1. The <i>BESF</i> shall maintain the set output level within $\pm 2\%$ of the capability registered with the <i>SO</i>, <i>NSP</i> or another <i>network operator</i> for at least one hour. 2. The <i>BESF</i> shall demonstrate ramp rates with precision within $\pm 2\%$ of the capability registered with the <i>SO</i>, <i>NSP</i> or another <i>network operator</i> for ramp up and down. 3. The <i>BESF</i> shall maintain a spinning reserve set level within $\pm 2\%$ of the capability registered with the <i>SO</i>, <i>NSP</i> or another <i>network operator</i> for at least one hour. 4. The <i>BESF</i> shall maintain an absolute power constraint set level within $\pm 2\%$ of the capability registered with the <i>System Operator</i> for at least one hour. 5. The <i>BESF</i> shall demonstrate that the requested frequency response curves can be obtained. Submit a report to the <i>SO</i>, <i>NSP</i> or another <i>network operator</i> one month after the test.

A2.2.4 - BESF reactive power control capability verification		
Parameter	Reference	Description
Reactive power control function and operational range	Sections 8 depending on category	APPLICABILITY All new <i>BESFs</i> coming on line and after major modifications or refurbishment of related plant components or functionality. Routine test/reviews: Confirm compliance every 6 years. PURPOSE To confirm that the reactive power control capability specified is met. PROCEDURE The following tests shall be performed within a minimum active power level range of at least 0.2 p.u. or higher 1. The <i>BESF</i> will be required to regulate the voltage at the PCC to a set level within the design margins. 2. The <i>BESF</i> will be required to provide a fixed Q to a set level within the design margins. 3. The <i>BESF</i> will be required to obtain a fixed PF within the design margins. ACCEPTANCE CRITERIA 1. The <i>BESF</i> shall maintain the set voltage within $\pm 5\%$ of the capability registered with the <i>SO</i>, <i>NSP</i> or another <i>network operator</i> for at least one hour. 2. The <i>BESF</i> shall maintain the set Q within $\pm 2\%$ of the capability registered with the <i>SO</i>, <i>NSP</i> or another <i>network operator</i> for at least one hour. 3. The <i>BESF</i> shall maintain the set PF within $\pm 2\%$ of the capability registered with the <i>SO</i>, <i>NSP</i> or another <i>network operator</i> for at least one hour. Submit a report to the <i>SO</i>, <i>NSP</i> or another <i>network operator</i> one month after the test.

A2.2.5 - BESF power quality calculations		
Parameter	Reference	Description
Power quality calculations for: 1. Rapid voltage changes 2. Flicker 3. Harmonics 4. Inter-harmonics 5. High frequency disturbances	Section 9 depending on category	APPLICABILITY All new <i>BESFs</i> coming on line and after major modifications or refurbishment of related plant components or functionality. Routine test/reviews: Confirm compliance every 6 years. PURPOSE To confirm that the limits for all power quality parameters specified is met. PROCEDURE The following tests shall be calculated within a minimum active power level range from 0.2p.u. to 1.0p.u. 1. Calculate the levels for rapid voltage changes are within the limits specified over the full operational range. 2. Calculate the flicker levels are within the limits specified over the full operational range. 3. Calculate the harmonics are within the limits specified over the full operational range. 4. Calculate the interharmonics are within the limits specified over the full operational range. 5. Calculate the disturbances higher than 2 Hz are within the limits specified over the full operational range. ACCEPTANCE CRITERIA 1. The calculations shall demonstrate that the levels for rapid voltage changes are within the limits specified over the full operational range. 2. The calculations shall demonstrate that the flicker levels are within the limits specified over the full operational range. 3. The calculations shall demonstrate that the harmonics are within the limits specified over the full operational range. 4. The calculations shall demonstrate that the interharmonics are within the limits specified over the full operational range. 5. The calculations shall demonstrate that the disturbances higher than 2 Hz are within the limits specified over the full operational range Submit a report to the <i>System Operator</i> one month after the test.

A.2.2.6 - BESF fault ride through simulations		
Parameter	Reference	Description
Simulations of fault ride through voltage droops and peaks.	Section 5. for category B and C	APPLICABILITY All new <i>BESFs</i> coming on line and after major modifications or refurbishment of related plant components or functionality. Routine test/reviews: None. PURPOSE To confirm that the limits for all power quality parameters specified is met. PROCEDURE By applying the electrical simulation model for the entire BESF it shall be demonstrated that the BESF performs to the specifications. 1. Area A - the BESF shall stay connected to the network and uphold normal production. 2. Area B - the BESF shall stay connected to the network. The BESF shall provide maximum voltage support by supplying a controlled amount of reactive power within the design framework offered by the technology, see Figure 4. 3. Area C - the BESF is allowed to disconnect. 4. Area D - the BESF shall stay connected. The BESF shall provide maximum voltage support by absorbing a controlled amount of reactive power within the design framework offered by the technology, see Figure 4. ACCEPTANCE CRITERIA 1. The dynamic simulations shall demonstrate that the BESF fulfils the requirements specified. Submit a report to the <i>SO, NSP or another network operator</i> three month after the commission.

Appendix 3 – Test Procedures for Gateway Factory Acceptance Tests (FAT), Site Acceptance Tests (SAT) and Commissioning

A4.1 Pre-FAT

(1) *NSP* or *SO* will provide the generic Factory Acceptance Test (FAT) procedures which shall be updated by the *BESF owner*. The final test procedure shall be agreed upon by the parties. The *BESF owner* shall submit the pre-factory acceptance tests results to *SO* or *NSP* at least two months prior to Grid connection date.

A4.2 FAT

(1) Factory Acceptance tests shall be done by the *BESF owner* where *NSP* or *SO* reserves the right to witness.

A4.3 Pre-SAT

(1) *NSP* or *SO* will provide generic Site Acceptance Test (SAT) procedures which shall be updated by the *BESF owner*. The final test procedure shall be agreed upon by the parties. Pre-site acceptance tests shall be done jointly by the parties using the *NSP* or *SO*'s applicable development systems to ensure inter-operability.

A4.4 SAT

(1) Site acceptance tests shall be done to the *NSP* or *SO*'s production systems.

A4.5 Commissioning

(1) Commissioning is the point to point testing of all specified signals to ensure that they are correctly configured and effect the correct operation.

(2) *NSP* or *SO* will provide test procedures for both plant online and offline testing. These procedures shall be used to verify the correct response to the issued commands.

Appendix 4 – Provision of Technical Network / Grid Data to *BESF Owners*

A4.1 Introduction

(1) *BESF Owners* that want to develop *BESF* plants in South Africa have to comply with the conditions set out in the applicable *Codes* which are approved by the *National Energy Regulator of South Africa (NERSA)*. The *NERSA Codes* define what technical grid information should be provided by the *Network Service Provider (NSP)* to the generators, and, the generators to the *NSP* for planning and design purposes. The aim of this appendix is to regulate provision of network/grid data from the *NSP* to the *BESF owners*. This document is applicable to all *BESF* connected to the medium voltage level and above. It is not applicable to *BESF* less than 1 MW.

A4.2 Information Exchange

A4.2.1 General

(1) The *NSPs* shall make every endeavour to provide the latest up to date information for the purposes required, however the *NSPs* do not guarantee the accuracy of the information. The information that shall be provided shall be the *NSPs* best available information at the time of request as stipulated in the *Code*.

A4.2.2 Simulation Studies Requirements for Generation Integration

A4.2.2.1 Studies to be performed by the NSP

(1) The *Network Service Provider (NSP)* is responsible to conduct generation grid impact studies. It must be noted that some of these studies are done in stages i.e. some are conducted during the feasibility (depending on generator technology types); while more detailed studies are only done during the design phases due to the overall grid impacts of all approved *Generators*.

(2) These studies are only required for *BESF* of category *B* and *C*.

(3) These studies may include:

- (a) Load flow studies, contingency analysis (verification of thermal loading and voltages),
- (b) Short Circuit studies (verification of short circuit rating of existing equipment, impact on protection),
- (c) Stability Studies:
 - (i) Transient stability studies (impact on critical fault clearing times, stability constraints and transfer limits of the entire connected system)
 - (ii) Oscillatory stability studies

- (iii) Frequency stability studies
 - (iv) Voltage stability studies (long-term and short-term)
- (d) Power Quality studies (The network harmonic impedance for network normal and a reasonable range of contingencies at the proposed point of connection).

A4.2.2.2 Studies to be performed by the *BESF Owner(s)*

- (1) The *BESF Owner* shall design and conduct grid code compliance studies.
- (2) The studies to be completed by the *BESF Owner* are as follows:
 - (a) Design studies
 - (i) Load flow and short circuit studies for design purposes (equipment rating)
 - (ii) Quality of supply (QoS) studies:
 - to meet allowed QoS emission limits
 - limit the resonance caused by the installation
 - anticipated voltage variations due to internal switching actions.
 - (b) Grid Code Compliance Studies
 - (i) Reactive power capability studies (based on load flows)
 - (ii) LVRT studies
 - (iii) HVRT studies
 - (iv) Power quality assessment (harmonics, voltage unbalance, rapid voltage changes and voltage flicker)

A4.2.3 Information Exchange between the NSP and the *BESF Owner*

- (1) *BESF* data exchange shall be a time-based process as follows.
 - (a) **First stage** (during the application for connection; also referred to as feasibility stage)
 - (i) The following information shall be submitted by the *BESF Owner* to the *NSP*:
 - Physical location of the generating plant (including the GPS coordinates)
 - Site Plan
 - Total generating plant MW output
 - Total capacity (MW) that will be exported into the grid of the NSP
 - Initial phase MW value
 - Final phase MW value and timelines
 - Any other information that the service provider may reasonably require

(ii) The *NSP* shall make available to the *BESF owner* or its duly appointed agent at least the following information:

- Define the *Point of Connection* and the *Point of Common Coupling* including the nominal voltages
- Expected maximum fault levels at the point of connection
- Expected minimum fault levels at the point of connection

(b) Second stage (during the design stage; for the *BESF Owner* to design his plant and conduct the grid code compliance studies).

(i) During this stage, the *NSP* shall provide information on the equivalent network as described in A4.2.4 below.

(ii) It must be noted that in certain cases, the development of an equivalent network will be time-consuming due to the numerous in-feeds and voltage levels that must be represented. This may also lead to instances where the accuracy of the equivalent network may be in question. Where no agreement can be reached on the equivalent network model developed by the *NSP*, then a full network model shall be utilized via joint studies by both the *NSP* and the applicant (*generator*).

A4.2.4 Data provisions by the NSP for the Equivalent Network, for the BESF design and grid code compliance studies (excluding harmonic studies)

(a) Source infeed

- (i) The source infeed(s) at the boundary bus(es) which represents the external network. This is typically one voltage level higher than the *POC* voltage.
- (ii) Transformer(s) connecting the boundary bus(es) to the surrounding network e.g. 400/132kV, 275/132kV, 132/33kV etc.
- (iii) Expected operating maximum and minimum fault levels [before and after connection, using the year of connection as the reference] at the boundary bus(es) (i.e. 400kV, 275kV, 132kV), including X/R ratios to model the correct network source in-feed(s) of the external network

(b) Fault levels and X/R ratios will be provided for the following network conditions:

- (i) Normal short circuit level,
- (ii) Maximum design short circuit level based on the planning horizon,
- (iii) Minimum short circuit level. Minimum short circuit level must be calculated on basis of an operational scenario with minimum generation (in the sourcing grid) in service (typically

minimum load case) and contingency scenario that gives the minimum fault level; provided the network is within an operable state.

Note: the calculation method for both the maximum and minimum short circuit levels shall be stated/included with the specific values i.e. IEC 60909.

- (iv) Worst case contingency scenario: Credible contingency scenarios (contingencies having a probability that is still in a range that makes it likely to be observed over a lifetime of e.g. 20 years) leading to minimum short circuit level/maximum network impedance at the POC.

(c) Surrounding network

- (i) Surrounding network at the same voltage level as the *POC*,
- (ii) The lumped peak load and minimum load (based on metering data, where available),
- (iii) If there is any generation within the surrounding network, such generation shall be represented by generic models,
- (iv) Any loads and/or generation at lower voltage levels, fed from the surrounding network will be lumped at the surrounding network voltage level,
- (v) Information on the available present network interruption performance and Quality of Supply (QOS) levels at the *PCC* or the closest node in the network; including the expected background harmonic distortion of the grid before connection of the generation facility. This shall include:
 - Indicative frequency sweeps at the *POC*
 - Voltage unbalance
 - Voltage dips
 - Network interruption performance data
 - Voltage Total Harmonic Distortion (THD) levels (where available)
 - The system normal and minimum fault levels at *POC*

Note: voltage unbalance, voltage dips, network interruption performance data, voltage harmonics is for information purposes only.

(d) The equivalent network

- (i) All lines (or cables) of the surrounding network. Note that the surrounding network is the same voltage of the *POC*.
- (ii) All synchronous power plants feeding into the surrounding network, using an aggregated generic model of the generating plant.
- (iii) All other non-synchronous power plants (e.g. wind farms, PV-power converters) using an aggregated, generic model (e.g. IEC-type model, IEEE standard models) at the POC of each of the farms.
- (iv) Loads shall be aggregated and modelled by one equivalent load connected to the surrounding network.

- (v) Feeders having load and generation shall be reduced and represented by one aggregated load and one equivalent generator connected to the surrounding network.
- (vi) Transformer(s) connecting to MTS of the surrounding network, e.g. 400kV/132kV, 275kV/132kV, 132kV/33kV
- (vii) Shunt compensation devices
- (viii) Source in-feed(s)

d1. Lines and cables data

- (i) Lines:
 - Length, positive and zero sequence impedance per length ($R1', X1' / R0', X0'$), positive and zero sequence capacitance (or susceptance per length) ($C1' / C0'$).
- (ii) Cables:
 - Length, positive and zero sequence impedance per length ($R1', X1' / R0', X0'$), positive and zero sequence capacitance (or susceptance per length) ($C1' / C0'$).

d2. Synchronous power plant data

- (i) Synchronous power plants having connection point in surrounding network:
 - Generator transformer (Rated capacity, short circuit impedance ($R1', X1', R0', X0'$), no-load impedance (iron losses/magnetizing reactance)
 - Synchronous generator (Rated capacity, voltage level and synchronous machine parameters)
 - Generic models for Automatic Voltage Regulator (AVR) and Power System Stabilizers (PSS)

d3. Non-synchronous power plant data

- (i) Other, Non-Synchronous power plants (e.g. wind farms, PV-farms) having connection point in surrounding network:
 - Aggregated generator transformer (Rated capacity, short circuit impedance ($R1', X1', R0', X0'$), no-load impedance (iron losses/magnetizing reactance)
 - Main MV bus-bar of generator installation
 - Aggregated, generic model of non-synchronous generator at main MV- bus bar of the windfarm/PV farm, e.g. IEC-type model/IEEE standard model.
- (ii) At a minimum, the following data/properties shall be represented:
 - Rating of the generation installation
 - Control mode during normal operation (farm controller, e.g. voltage, power factor etc.)

- Reactive power capacity (using grid code characteristic)
- Dynamic model with main focus on the generator's behaviour during LVRT-events.
- If additional dynamic components are used (e.g. STATCOM, Capacitors, SVCs) must also be represented

d4. Feeders

- (i) Feeder loads:
 - Lumped, aggregated load model (without feeder transformer).
 - Maximum load with applicable power factor.
- (ii) Feeders with embedded generation:
 - Parallel connection of equivalent Feeder-Load and equivalent, generic generator.
 - Maximum load with applicable power factor.
 - Maximum generation with operating power factor

d5. Transformers and shunts

- (i) MTS Transformer (e.g. 400/132kV, 275/132kV, 132/33kV):
 - Transformer rating
 - Rated voltages of transformer
 - Short circuit impedance (R, X)
 - No-load current and iron losses
 - Information about tap changer (number of taps, tap steps, max./min. taps, on-load/off-load, controlled node, control mode)
- (ii) Shunt compensation devices
 - Generic models of the shunt compensation devices
 - All the voltage switching ranges
 - Control set point

A4.3.5 Examples

- (a) Connection to a sub-transmission network with a single source in-feed:
 - (i) Proposed generating plant POC is at 132kV as shown in 0.

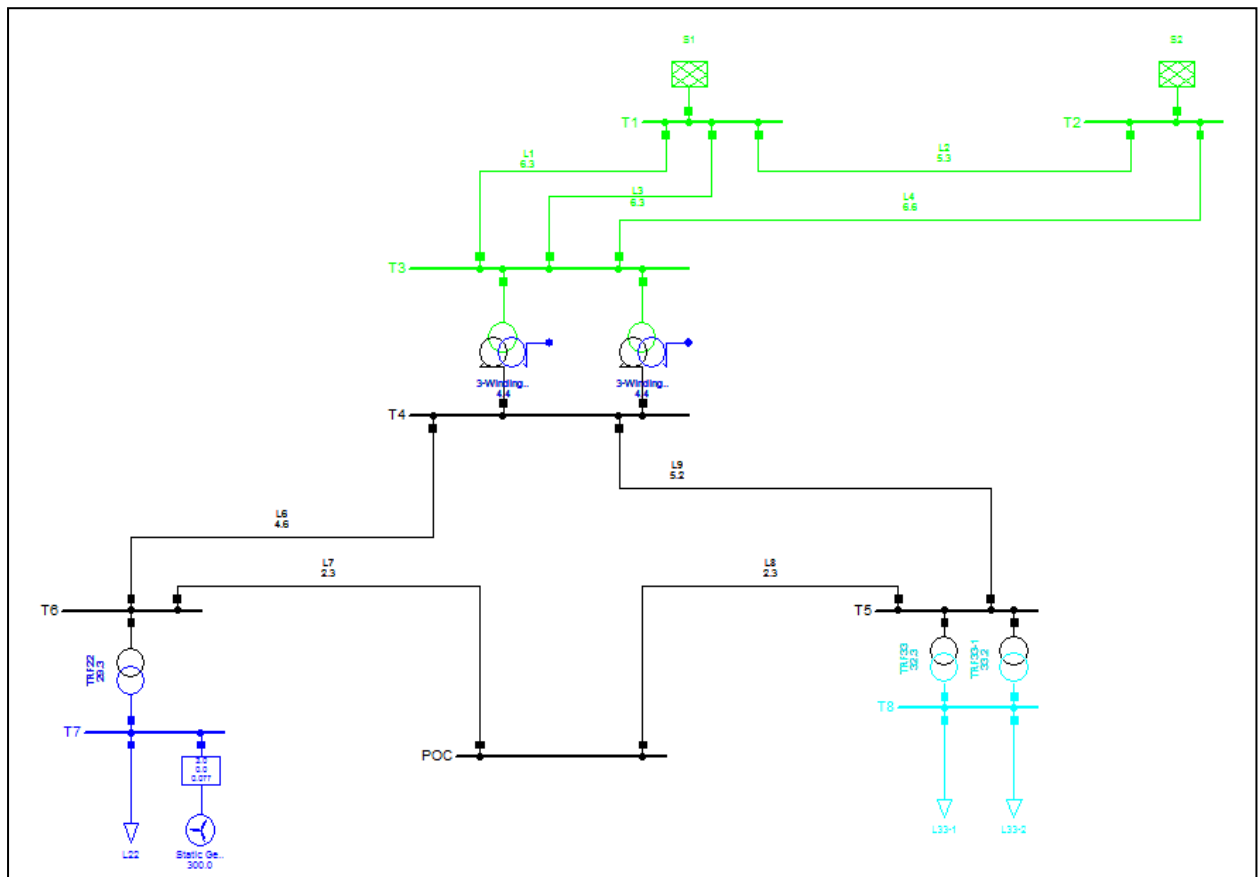


Figure A4-1: Network showing a single source in-feed and the proposed connection PoC is at 132kV

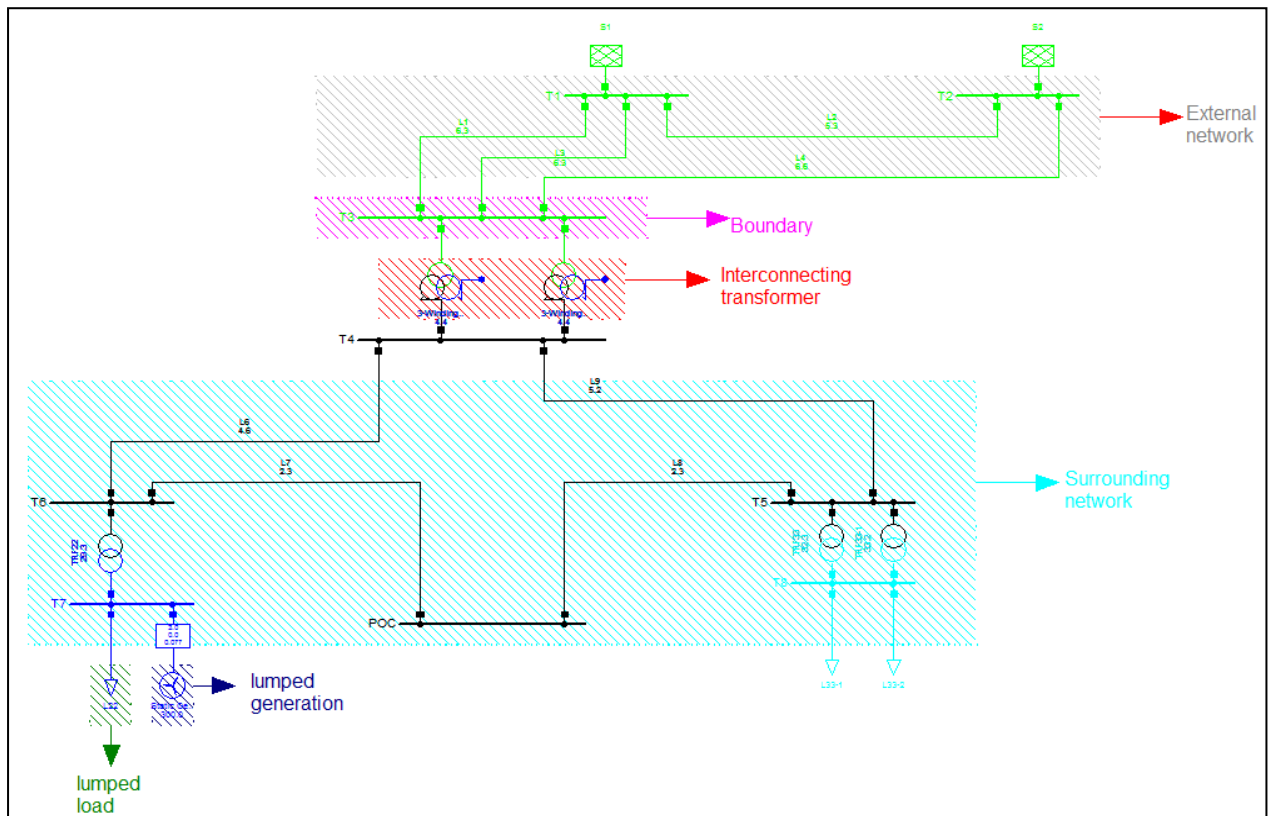


Figure A4-2: Single source in-feed network illustrating some definitions

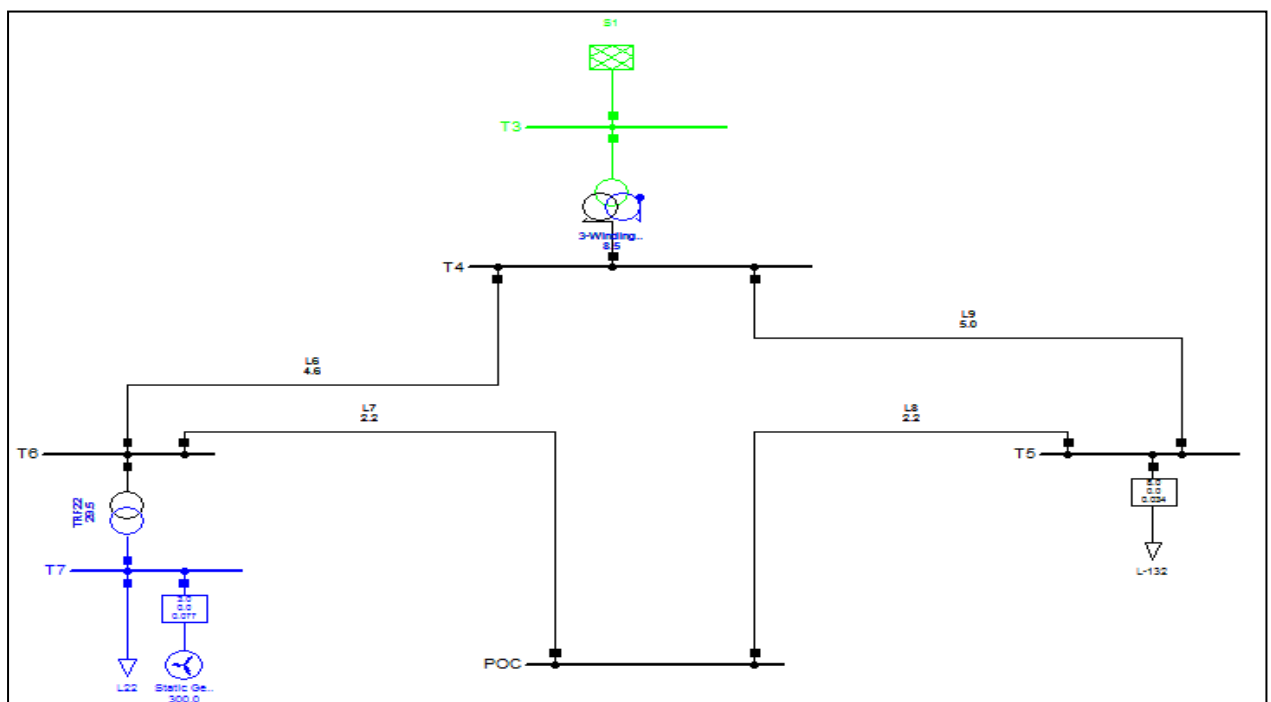


Figure A4-3: Equivalent network with single source in feed

(b) Connection to a sub-transmission network with two source in-feeds:

(i) Proposed generating plant is at 132kV as shown in 0.

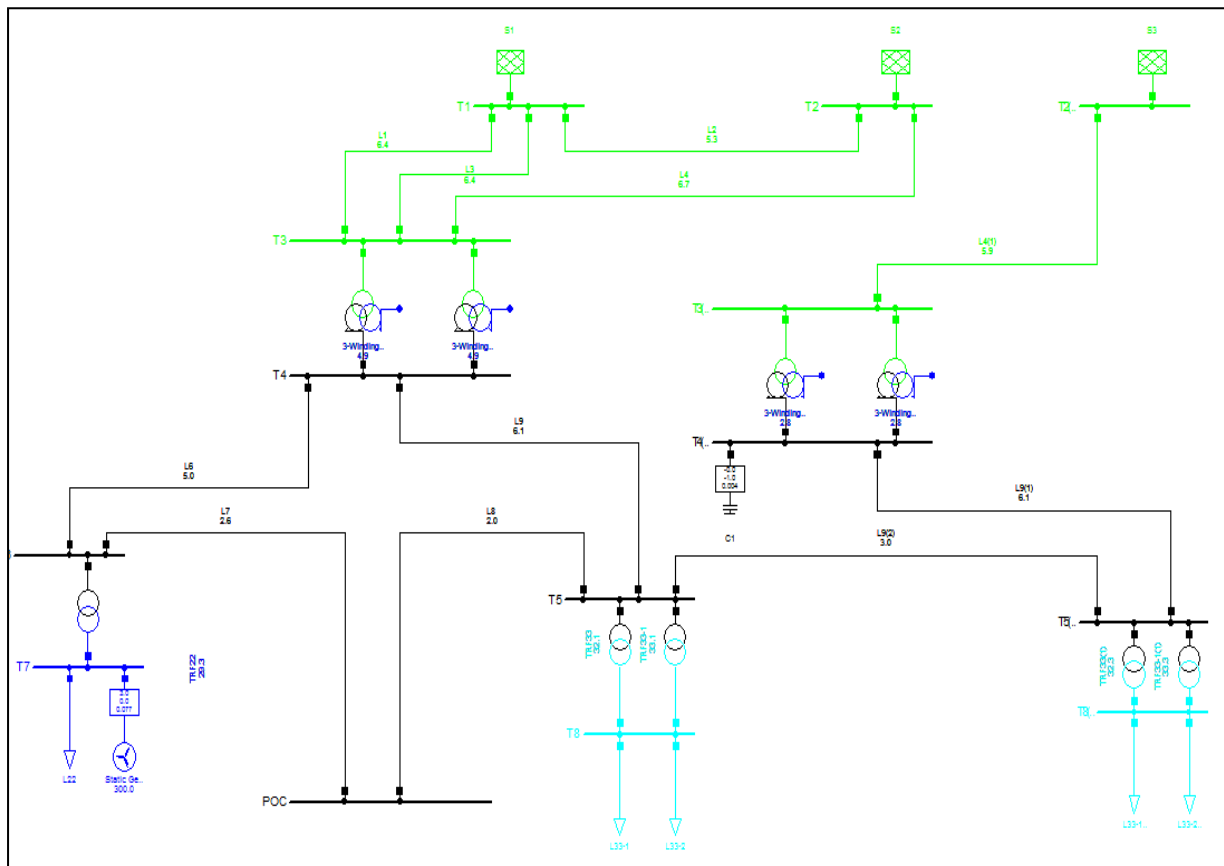


Figure A4-4: Network with two source in-feeds showing the POC

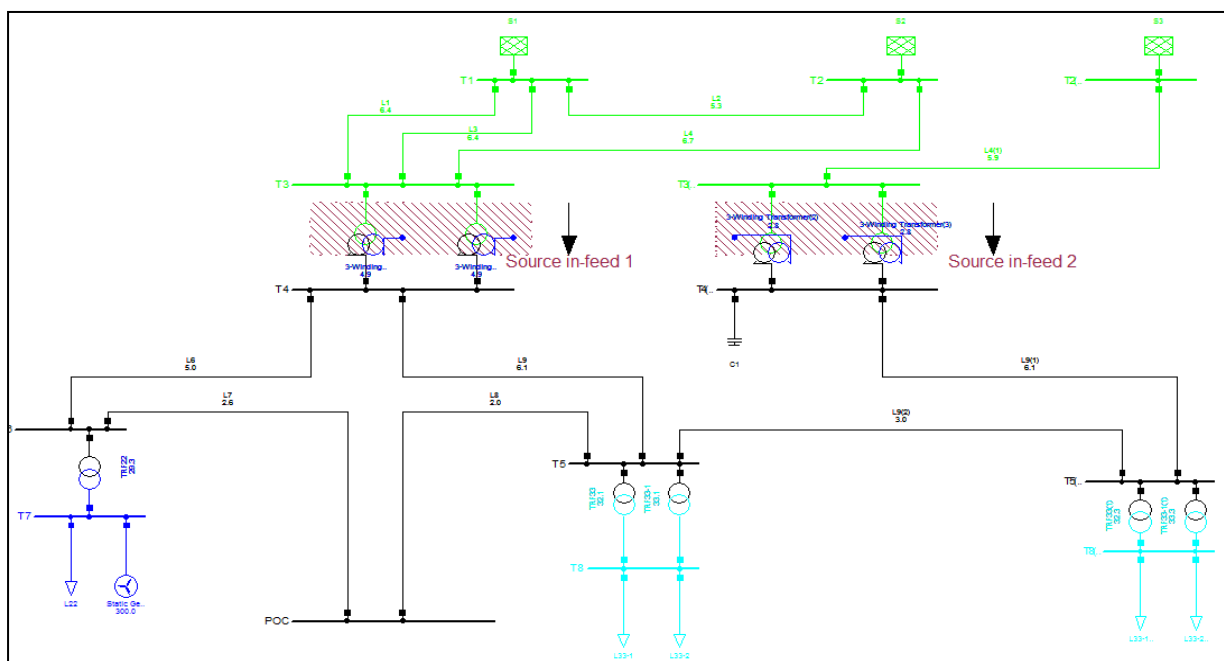


Figure A4-5: Two source in-feeds network illustrating some definitions

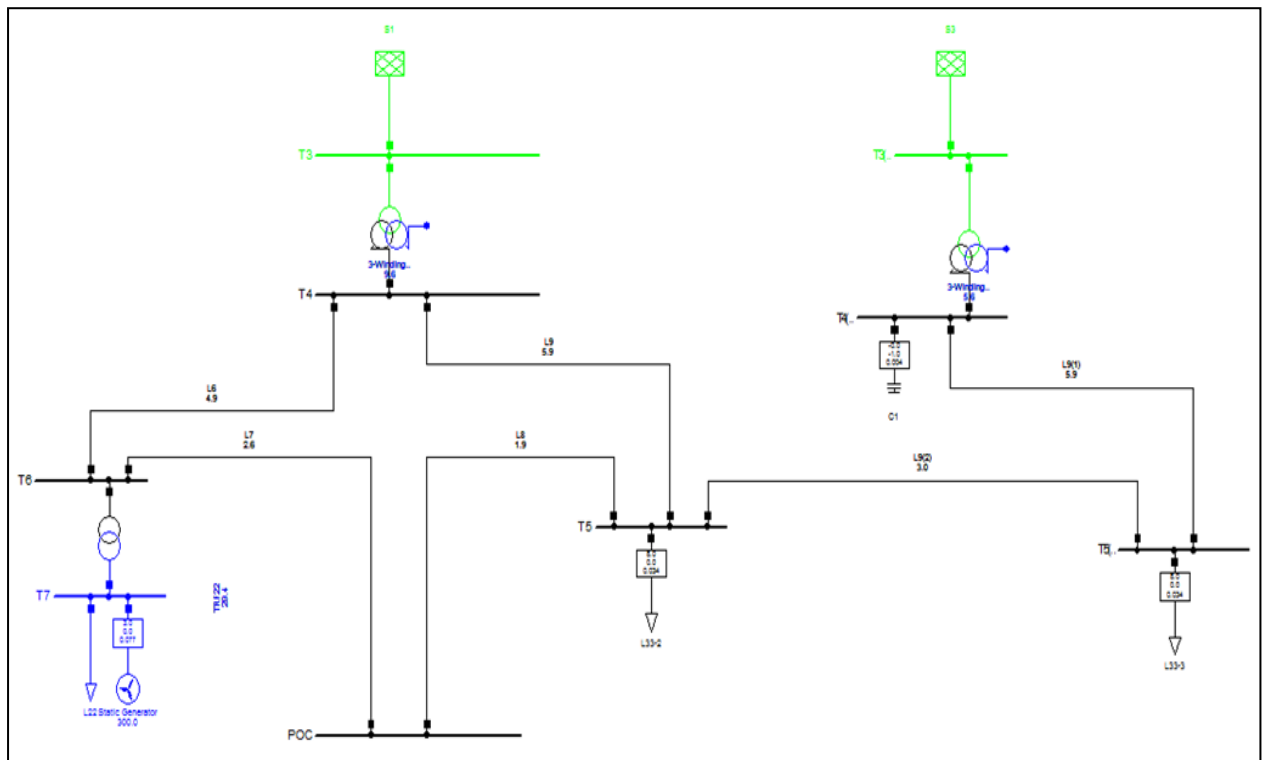


Figure A4-6: Equivalent network of two source in-feeds

Appendix 5 – BESF Power Quality Compliance Guideline

A5.1 Introduction

- (1) This document provides requirements for the compliance process to enable *BESF Owner(s)* to meet the Grid Code requirements with respect to power quality (PQ) as stipulated by the NSP.

A5.1.1 Definition of Emission Level:

- (1) The definitions as stipulated in the IEC 61000-3-6 document for harmonic emission level is interpreted as the magnitude of the vector that is caused by the installation at the point of evaluation (POC for *BESF*). The concept is illustrated in Figure A5-1 for a voltage harmonic phasor.

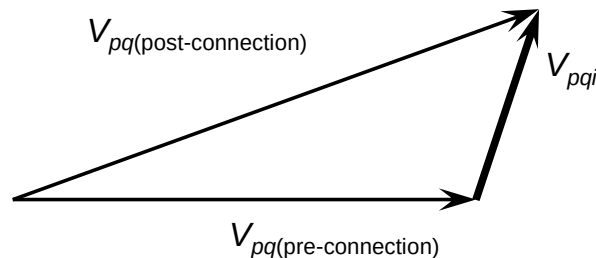


Figure A5-1: Definition of emission level.

- (2) A similar definition is provided for voltage unbalance in IEC 61000-3-13. Whilst the definition is not as clear for voltage flicker in IEC 61000-3-7, the principle is similar, since voltage flicker would not add linearly between different installations unless there is an intentional link.

A5.2 Power Quality Emission Limits

- (1) NSP's shall calculate the specific emission limits as guided by NRS048:4 or IEC 61000-3-6.

NOTE: NRS048:2 do not specify emission limits but rather compatibility levels that the NSP must account for. Compatibility levels as stated in NRS048:2 shall not be issued to Generators as emission levels.

A5.3 NSP and Reference Data

- (1) In line with the provision of technical data guideline, the following information is relevant to the *BESF Owner* for power quality purposes (where available):
 - (i) If available, historical performance of power quality background levels at the PoC or closest available point; as plots of the 95th percentile weekly values (10 minute values) for the voltage trends for at least a year or the period for which data is available.

- (ii) A range of impedance frequency sweep data at the *POC*, including network healthy and a realistic range of network contingencies.

NOTE: These contingencies are based on network operational requirements in consultation between the *NSP* PQ engineer and *NSP* Network Operational personnel.

- (iii) A range of impedance frequency sweep data at the *POC* for the future network. This data may be used to assess the future harmonic impact due to varying impedance, however it would not be any guarantee that the network would evolve as provided.

NOTE Frequency sweep data should be provided as the complex network impedance vs. frequency, i.e. R and X , or, Z and ϕ ,

- (2) Reference data includes compatibility levels for all power quality parameters as provided in NRS048-2, or as specified in the connection agreement.
- (3) It is important to note the interaction between the various power quality parameters; e.g. background harmonics and voltage unbalance that often impact the harmonic emissions from equipment.

NB: Background levels/total levels after connection of all power quality parameters up to the compatibility levels should be taken into account when calculating emission levels as well as immunity of equipment. The historical background data is, therefore, not critical for any power quality assessment.

A5.4 Compliance Assessment

- (1) The main compliance criteria for harmonics will be harmonic current measurements at the PoC (Although both voltage and current harmonic emission levels are prescribed by the NSP).
- (2) Compliance for voltage flicker shall be based on measured voltage values at the PoC.
- (3) Compliance for voltage unbalance shall be based on measured voltage or current (negative sequence current) values at the PoC.
- (4) Grid code compliance must be proven by means of measurements under normal operating conditions (i.e. *BESF* operating) for a period of no less than 7 consecutive days (24 hours per day). Measurements should be taken once the Generator is connected to the grid but is not yet commercially operational. The generation profile must be representative of the *BESF* final commercial operations, i.e. with all generation capacity in service.
- (5) Harmonic emission simulations will not be required for grid code compliance, however should the;

- (i) NPS experience a violation of the PQ levels within NRS048:2 due to the *BESF* connection, the NSP must request that the *BESF Owner* to reduce its output within 7 days so as to manage the PQ levels to within NRS048:2 compatibility levels.
- (ii) *BESF Owner* become aware or should be reasonably being aware of a violation of the PQ levels within NRS048:2, the *BESF Owner* must reduce its output within 7 days to manage the PQ levels within NRS048:2 compatibility levels.

A5.4.1 Assessment Process

A5.4.1.1 Generators larger than 5 MVA (Category B >5 MVA and larger categories)

(1) The following assessment process is applicable to all *BESF* larger than 5 MVA:

STEP 1: *BESF Owner* to obtain voltage and current emission limits from the NSP.

STEP 2: *BESF Owner* to measure and report power quality assessment, in line with prescribed methods in section A5.5, to the NSP within 6 months of grid connection (note that the plant generation pattern must be representative as if the plant is in full commercial operation). Measurements must include a period (at least 24 hours continuous) where the *BESF* is not connected (connecting breaker open) for the harmonic, unbalance and flicker assessment, these measurements must be taken at the PoC prior to the plants.

STEP 3: *BESF Owner* to engage with the NSP and discuss the results of the emission assessment. Where violations of the respective harmonic emission limits are evident, agreement must be reached on which harmonics are emitted and which are absorbed/sunk by the *BESF*. The same holds true for voltage unbalance and voltage flicker.

Where there are no emission violations then the *BESF* will be considered to be Grid Code compliant for PQ.

Where there are violations of the emission levels and if agreement is obtained between the NSP and *BESF* on the harmonic emission violations i.e. absorbed vs emitted, then the NSP will support an appropriate conditional exemption for the *BESF* to mitigate the emission violations.

NOTE: Clause A5.4(5) applies during this exemption period.

The above mentioned process holds true for voltage unbalance and voltage flicker.

Step 4: *BESF Owner* must redo their compliance assessment after mitigation, within exemption period granted and present to NSP for final grid code approval. (The 24 hour period of disconnection may not be required if all emission levels are below the allocated emission limits.)

Figure A5. 2 shows a flows diagram of the assessment process described above.

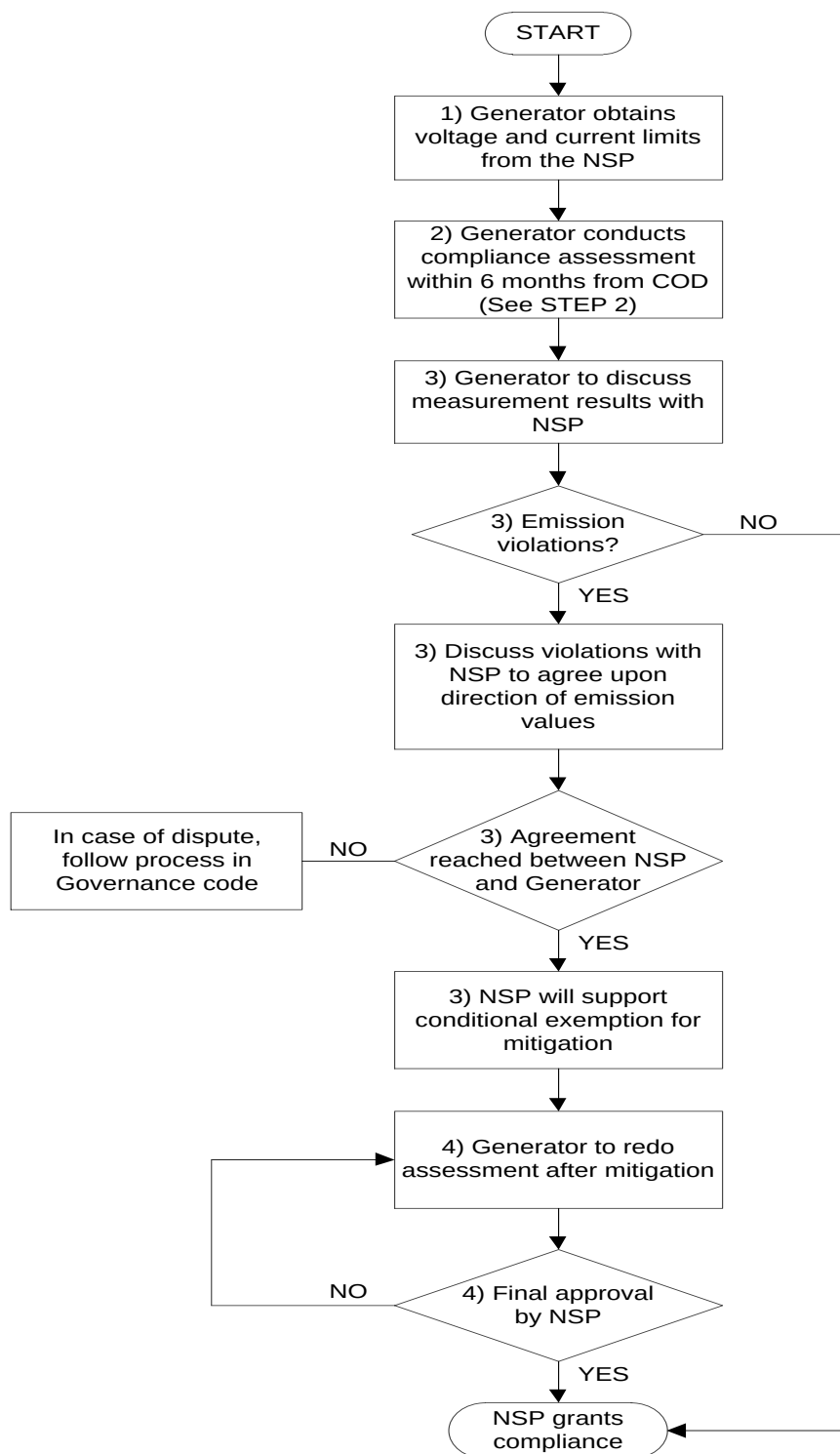


Figure A5-2: Flow diagram of assessment process

A5.4.2 Resonance Screening Process

- (1) The basic screening process involves checking whether the connection of the *BESF* will cause excessive resonance. Resonance is regarded as excessive when the impedance envelope ($3 \times Z_h$) is exceeded at any frequency.
- (2) *BESF Owner(s)* are encouraged to engage with the *NSP* for the resonance screening. *BESF Owner(s)* are to integrate their plant models into the *NSP* utility model (DigSilent simulation tool) to evaluate the combined impact on the resonance screening. Only the results (frequency sweeps) of such integration will be shared with the *BESF*. Full *NSP* models will not be shared.

A5.4.3 Emission Requirements for Power converters used in Category A and B ($\leq 5\text{MVA}$) Generator

A5.4.3.1 Harmonics

- (1) The individual harmonic currents of the power converters shall not exceed the limits specified IEC61727. These values are specified in Table 2 and the total harmonic distortion (THD) of the current (to the 50th harmonic) shall be less than 5%.

Table A5.4.3-1: Inverter harmonic emission requirements

Harmonic order (h)*	$3 \leq h < 9$	$11 \leq h < 15$	$17 \leq h < 21$	$23 \leq h < 33$	$33 \leq h \leq 50$	Even	THD
Percentage of rated current (Odd harmonics)	<4,0	<2,0	<1,5	<0,6	<0,3	25% of odd harmonics	<5,0

A5.4.3.2 Flicker

- (1) The inverter shall conform to the voltage fluctuation and flicker limits as per SANS61000-3-3 for equipment rated less than or equal to 16 A per phase and SANS61000-3-5 for equipment rated greater than 16 A per phase. Compliance shall be determined by type testing in accordance with the appropriate Standard.

A5.4.3.3 Voltage Unbalance

- (1) For *BESF* connected to multi-phase supplies (two- or three-phase connection at the POC), the difference in installed capacity between phases may not exceed 4.6 kVA per phase

A5.5 Grid Code Compliance: Measurement

A5.5.1 Measurement Requirements

- (1) All *BESF* above $>5\text{MVA}$ are required to demonstrate compliance at installation level via power quality measurements. All measurements for compliance purposes will be done at the PoC, unless not practically feasible. Voltage and Current measurements shall be taken.

- (2) The following requirements apply for these measurements:
- The PQ instrument(s) shall comply with IEC 61000-4-30, class A Edition 3 (or latest version).
 - The instrument(s) shall be set-up to record the 10-minute aggregated values as a minimum. Where possible phasor measurements (including aggregated phasors) may be measured and applied.
 - These instruments shall be set-up to record all current and voltage harmonics up to the 50th harmonic.
 - Three-phase measurements shall be done:
 - For a *PoC* that is connected in Wye with a solidly earthed neutral, the phase-to-neutral voltages and the phase currents shall be monitored;
 - For all other systems, the phase-to-phase voltages and the phase currents shall be monitored.
 - Any deviation shall be motivated and confirmed with the *NSP* when required.
 - Where the *PoC* has two or more feeders to the *BESF*, a physical current summation method shall be implemented.
- (3) Measurements used for the analysis of the emissions shall be taken for a minimum of one week, starting at midnight on day 1 lasting until midnight after 7 days. The duration of the measurement shall be representative of the plant's expected normal production cycle.
- (4) Simultaneous, GPS synchronised measurements must be taken when more than one PQ instrument is used in the assessment of Grid Code Compliance.
- (5) A measurement uncertainty assessment must be completed for the measurement circuit utilised by the Generator, these include all instrument transformers, transducers, cable lengths and the measurement instrument. The JCGM 100 and JCGM 106 documents may be used to calculate the measurement uncertainty as well as to apply the uncertainty to the measured parameters. Accuracy levels for the equipment utilised are specified in the equipment specification sheets. Generic accuracy levels for instrument transformers may also be obtained from the IEC 61000-4-7 and from IEC 61869-103. The measurement uncertainty assessment shall be included in the assessment report to the *NSP*.

A5.5.2 Harmonic Emission Evaluation/Assessment for category B (>5MVA) and C Generator's

- (1) There are cases where the current flowing through the *PoC* will exceed the limits even though the emissions of the plant itself do not exceed these limits. This will be true where the plant presents low impedance at a certain frequency and therefore absorbs harmonic currents from the network and acts as a harmonic filter. For these cases, and others, a number of different methods (as listed below) can be used to prove that the *BESF* is absorbing the harmonic currents measured and not

emitting them. The *BESF* must remove the measured values deemed to be absorption and prove compliance based on the remaining dataset (considered to be emission only). The 95th percentile value of the emission dataset must be compared to the respective limit to prove compliance.

(2) The different methods include the following:

- a) The impedance slope method, as described in Cigré Technical Report 468: Review of Disturbance Emission Assessment Techniques, section 4.2.2: Harmonic Emissions Level Compliance Assessment;
 - Use only if the linear trend line of the 10 minute measurements has a positive slope
 - Impedance slope must reflect the network at the time of measurement
 - The load impedance slope must represent the full Generator internal network.
- b) Consistent reduction background voltage harmonic emissions coinciding with the increase of measured harmonic currents during power generation by the Generator.
- c) Measurement and assessment of harmonic emission using harmonic phasors, (including aggregated harmonic phasors).

(3) Triplen harmonics, in a Delta/Star or Star/Delta transformer system, may be equated to zero for balanced systems (i.e. zero negative phase sequence measured) only. For all other unbalanced systems the following equation will apply to cater for the penetration of triplen harmonics through the Delta windings.

$$\%I_h = 0.95h * V_{\%UB}$$

Where $\%I_h$ is the percentage of harmonic current flowing through the star-delta transformer, h the harmonic order and $V_{\%UB}$ the percentage voltage unbalance measured.

A5.5.2.1 Relaxation of Limits: Harmonic Current Emission Evaluation for Category B (>5MVA) and C Generators

(1) With respect to the requirements for meeting harmonic emission limits provided by the *NSP*, the *BESF Owner* shall be allowed to exceed individual current harmonic emission limits by up to 50% (e.g. if the 5th harmonic limit is 1A, the *BESF* may emit up to 1.5A) provided that the Harmonic Distortion (HD) band limit is met for the following specified bands: $2 \leq h \leq 13$, $14 \leq h \leq 25$, $26 \leq h \leq 39$, $40 \leq h \leq 50$.

The THD bands are defined as follows:

$2 \leq h \leq 13$:

$$HD_{LimitG1} = \sqrt{\sum_{h=2}^{13} I_{limit}(h)^2} \quad HD_{MeasuredG1} = \sqrt{\sum_{h=2}^{13} I_{measured}(h)^2}$$

Grid Code compliance requirement: $HD_{MeasuredG1} < HD_{LimitG1}$

For $14 \leq h < 25$:

$$HD_{IlimitG2} = \sqrt{\sum_{h=14}^{25} I_{limit}(h)^2} \quad HD_{I_{measured}G2} = \sqrt{\sum_{h=14}^{25} I_{measured}(h)^2}$$

Grid Code compliance requirement: $HD_{I_{measured}G2} < THD_{IlimitG2}$

For $26 \leq h \leq 39$:

$$HD_{IlimitG3} = \sqrt{\sum_{h=26}^{39} I_{limit}(h)^2} \quad HD_{I_{measured}G3} = \sqrt{\sum_{h=26}^{39} I_{measured}(h)^2}$$

Grid Code compliance requirement: $HD_{I_{measured}G3} < THD_{IlimitG3}$

For $40 \leq h \leq 50$:

$$HD_{IlimitG4} = \sqrt{\sum_{h=40}^{50} I_{limit}(h)^2} \quad HD_{I_{measured}G4} = \sqrt{\sum_{h=40}^{50} I_{measured}(h)^2}$$

Grid Code compliance requirement: $HD_{I_{measured}G4} < HD_{IlimitG4}$

- (2) Where it has been shown that a plant absorbs a specific harmonic order, this harmonic order may be reduced to the allowable limit (as allocated by the NSP) in the Group HD band calculation.
- (3) The HD Band requirement is illustrated in figure A5.5.2-1 for clarity.

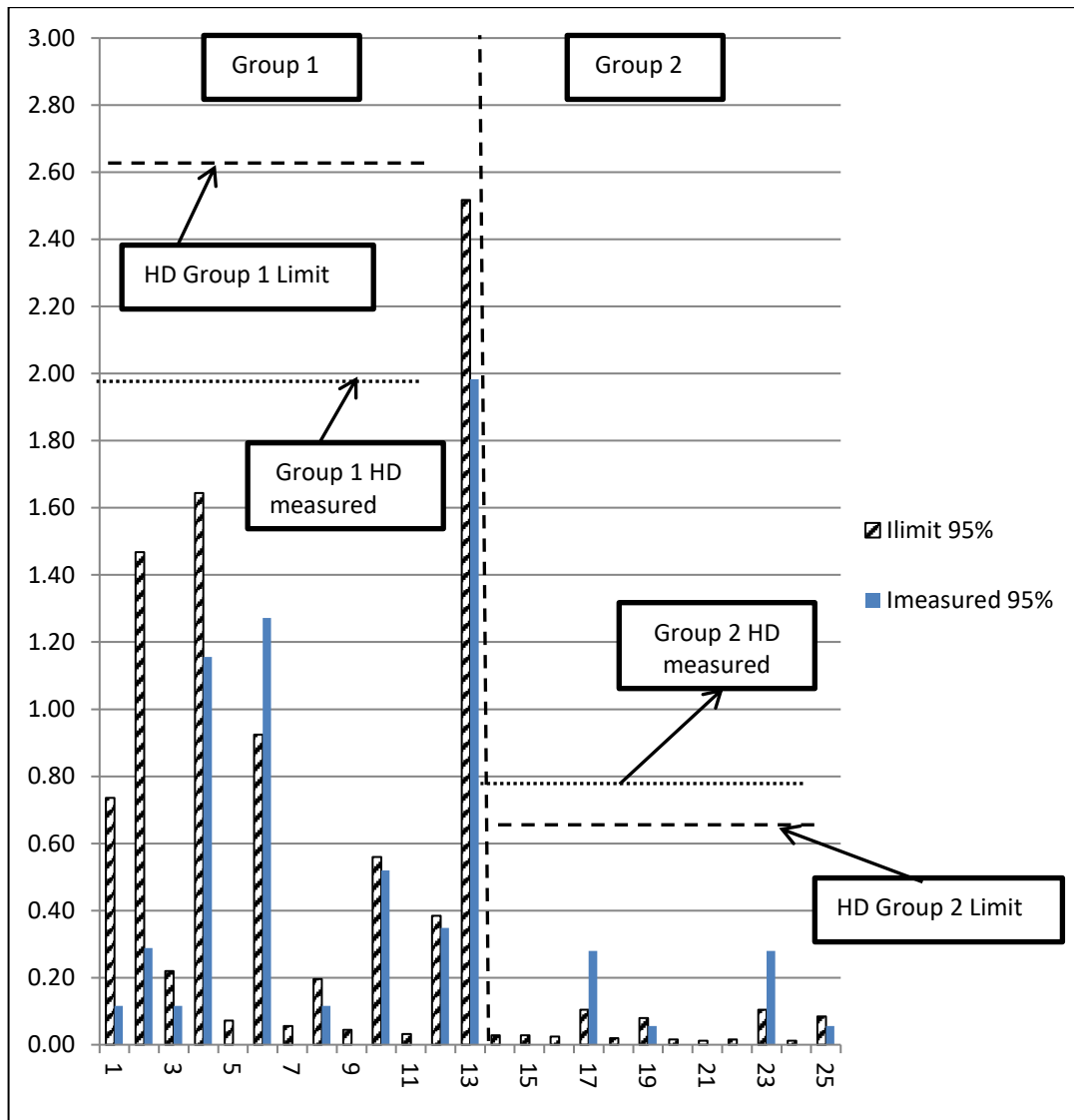


Figure A5.5.2-1: Group HD Emissions Assessment

- (4) Figure A5.5.2-1 shows the individual current harmonic measurements and limits as well as the HD bands measurements and limits for bands $2 \leq h \leq 13$ (Group 1) and $14 \leq h \leq 25$ (Group 2).
- (5) For HD band $2 \leq h \leq 13$, it can be observed that the measured current for the 7th harmonic exceeds the limit, however the Group 1 HD measured is less than the limit set for the HD Group. The *BESF* is deemed to meet the harmonic emissions requirements and does not need to take any actions.
- (6) For HD band $14 \leq h \leq 25$, it can be observed that the measured harmonic currents for the 17th and the 23rd harmonic exceed their limits. Additionally, the Group 2 HD measured exceeds the Group 2 HD limit. In this instance the *BESF* is required to take actions to ensure that the Group 2 HD harmonic emissions are reduced to below the limits.

- (7) The weekly sliding 95% values at each harmonic current shall be assessed against the emission limits specified by the *NSP* and the values which were calculated.

A5.5.3 Flicker Emission Evaluation

- (1) The following method is recommended for the assessment of flicker.
- (2) This assessment method for flicker is based on the method from the Cigré Technical Report 468: Review of Disturbance Emission Assessment Techniques, section 5.3.2 - Simplified Approach. The method has been modified to reflect the requirements for Plt as specified in South African standards.
- (3) However, any of the other methods listed in section 5.3 of Cigré Technical Report 468 may also be used by the Generator to assess flicker.
- (4) The typical configuration for individual flicker emission level assessment is given in Figure A6.5.3-1. The installation under consideration is connected at point A. This point is connected to the *POC* (point B) through impedance $\bar{Z}_2$ (in this case the transformer connecting to the *NSP*). Other fluctuating loads are possibly connected at B.
- (5) Typically, when other fluctuating loads are operating in the electrical vicinity, the background flicker at the *POC* (point B) cannot be neglected. However, on the secondary side of the transformer (A) the dominance of the emission Plt emission (A) of the investigated installation in the global flicker level Plt (A) increases and the influence of other sources can often be neglected [13], especially when the 95th on 99th percentile of Plt are considered (IEC 61000-3-7).

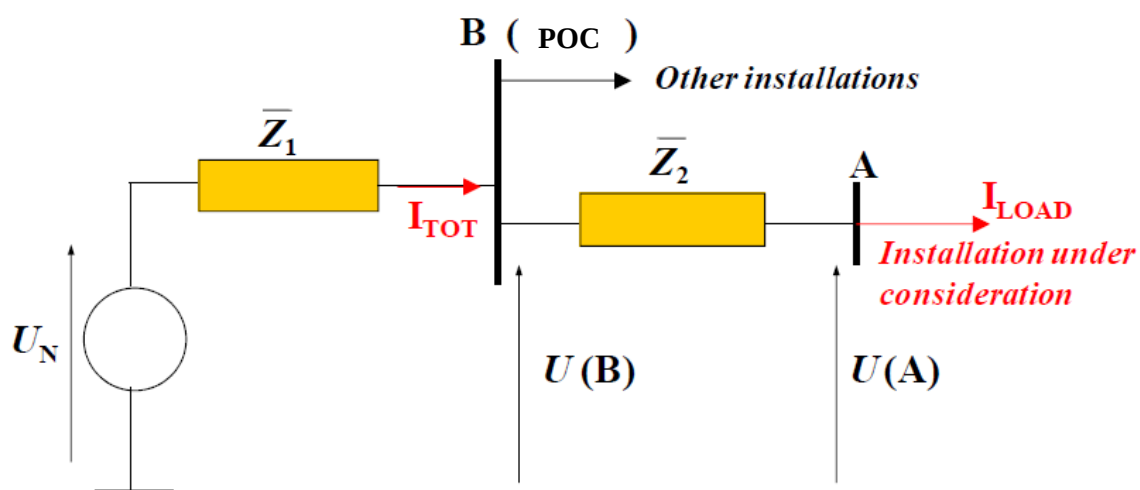


Figure A5.5.3-1: Network configuration for the application of the "simplified approach"

- (6) The simplified approach consists of measuring the flicker level on the secondary side(A) of the transformer and to transpose it to the primary side (B), by simply using the impedance ratio below.

$$P_{lt\ emission}(A) \cong P_{lt}(A)$$

$$P_{lt\ emission}(B) \cong \left| \frac{\bar{Z}_1}{\bar{Z}_1 + \bar{Z}_2} \right| P_{lt\ emission}(A) = \frac{S_{SC,A}}{S_{SC,B}} P_{lt\ emission}(A)$$

The following definitions apply:

- Z_1 is the network impedance,
- Z_2 is the transformer impedance,
- $P_{lt}(A)$ is the global flicker level at A,
- $P_{lt\ emission}(A)$ is the flicker emission level of the installation under consideration, at A,
- $P_{lt\ emission}(B)$ is the flicker emission level of the installation under consideration, at B (POC),
- $S_{SC,A}$ is the short-circuit power at A and
- $S_{SC,B}$ is the short-circuit power at B (POC).

- (7) The short circuit power at the POC, $S_{SC,B}$ shall be provided by the NSP.
- (8) All flicker measurements used shall be the Weekly 95th percentile of the voltage flicker Plt and Pst

A5.5.4 Voltage Unbalance Emission Evaluation/Assessment

- (1) The following method is recommended for the assessment of voltage unbalance:
- (i) The assessment for voltage unbalance is taken from the Cigré Technical Report 468: Review of Disturbance Emission Assessment Techniques, section 6.4.3.3 – Emission level Assessment based on Current Measurement.
- (2) However, any of the other methods listed in section 6.4.3 of Cigré Technical Report 468 may also be used by the Generator to assess voltage unbalance.

A5.5.4.1 Total emission level assessment based on measurement before and after connection

- (1) Connection of an installation to the system can lead to an increase or decrease (see Figure A6.5.4-1b) of the resultant unbalance level (i.e. $U_{2,i}$). This can be investigated by measuring the unbalance levels at the POE before (i.e. the unbalance level = $U_{2,oc}$ and labelled as $U_{2,pre-connection}$) and after (i.e. the unbalance level = U_2 and labelled as $U_{2,post-connection}$) the connection. The increase or decrease of the resultant unbalance level is determined by the phase angle of the emission $U_{2,i}$ caused by the connection of the load.

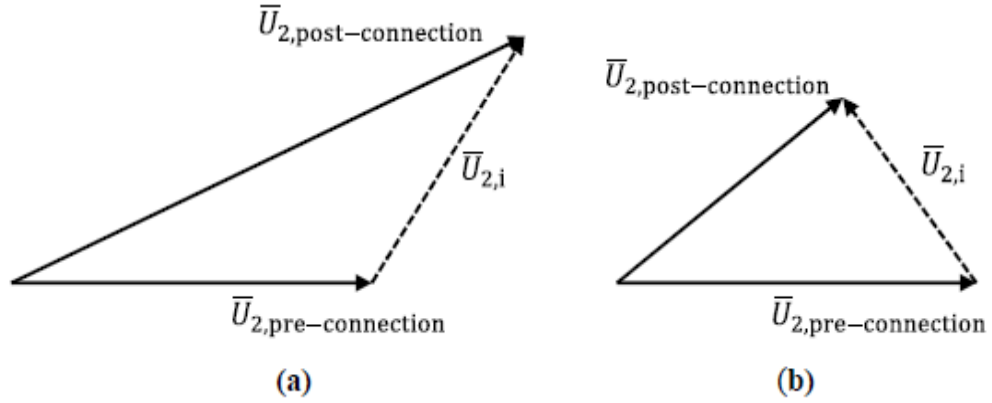


Figure A5.5.4-1: Comparison of the voltage unbalance level before and after the connection of an installation

(2) This is the first step which leads to one of the following procedures:

- If a decrease of the resultant unbalance level is caused by the connection of the installation, no emission limit is allocated to the installation and no further action is necessary.
- If an increase of the resultant unbalance level is caused by the connection of the installation, the fraction of the emission level which the installation is responsible for (i.e. $U_{2,i\text{-load}}$) has to be determined for the purpose of comparison against the emission limit allocated based

(3) In the case of measuring instruments providing the phase angle information and when it is known that the vectors $U_{2,\text{pre-connection}}$ and $U_{2,\text{pre-connection}}$ are likely to be in-phase, the total emission $U_{2,i}$ caused by the connection of the load can be determined by:

$$|\bar{U}_{2,i}| = |\bar{U}_{2,\text{post-connection}} - \bar{U}_{2,\text{pre-connection}}|$$

(4) In the case where the above information is unavailable, as directed in IEC/TR 61000-3-13, the summation law ($\alpha = 1.4$) can be applied to determine the emission $U_{2,i} = |U_{2,i}|$ as shown in:

$$|\bar{U}_{2,i}| = \left(|\bar{U}_{2,\text{post-connection}}|^\alpha - |\bar{U}_{2,\text{pre-connection}}|^\alpha \right)^{1/\alpha}$$

(5) All unbalance measurements used shall be Weekly 95th percentile of the unbalance measured on a 10-min basis.

Appendix 6 – Calculation of Errors for Validation of EMT Models

A6.1 Introduction

(1) The accuracy of *EMT* model representation shall be assessed during model validation. The validation shall provide comparison of recorded oscillographies of voltages and currents at *POC* to simulated values of the same voltages and currents during fault event external to the *BESF*.

(2) Evaluated currents shall represent sole contribution of the *BESF* to the fault on the *NSP* network.

(3) The severity of the event shall be such that the voltage depression at *POC* shall exceed 20% of *Un*.

A6.2 Validation Criteria

(1) The following requirements shall be met for successful validation of the model:

1. Voltage validation requirements:
 - a. Average absolute error of sample-by-sample evaluation shall not exceed 5% in all 3 phases;
 - b. The 90% error band in sample-by-sample evaluation shall not exceed $\pm 10\%$ in all 3 phases. Where deviations occur, these shall be explained based on sound engineering principles.
2. Current validation requirements:
 - a. Average absolute error of sample-by-sample evaluation shall not exceed 15% in all 3 phases;
 - b. The 90% error band in sample-by-sample evaluation must not exceed $\pm 20\%$ in all 3 phases. Where deviations occur, these Transgressions shall be explained based on sound engineering principles;
 - c. Average error of 1 cycle RMS values shall not exceed 15% in all 3 phases;
 - d. Average error of Symmetrical Components values shall not exceed 20% for all 3 Symmetrical Components (positive, negative and zero sequence). Calculations shall be done using 1 cycle Fourier Transform.

- (2) Calculations of the above indices shall be done over the fault duration and include one 50Hz cycle after the fault was cleared.
- (3) For RMS and Sequence Components calculations the front of the single cycle window shall start one cycle after fault inception and end 2 cycles after the fault was cleared.
- (4) Comparisons for voltages and currents as well as calculated errors shall be presented graphically, on time axis, to reflect pre-fault conditions, fault duration and post-fault conditions.
- (5) Calculation spreadsheets shall be submitted in support of verification of results.

A6.3 Examples of Graphical Presentation

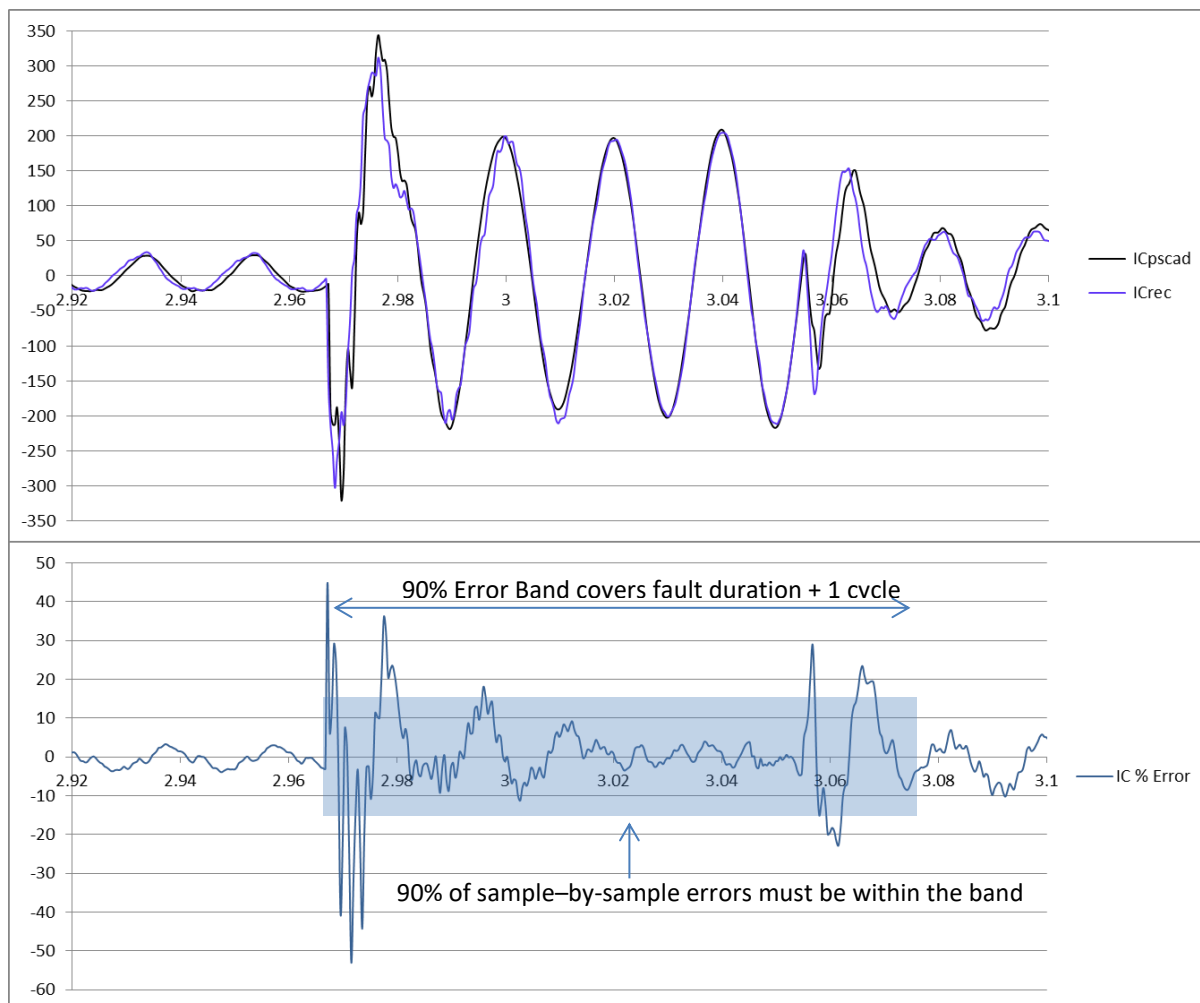


Figure A6.3.1 An example of superimposed simulated and recorded currents and sample-by-sample errors with error band over fault duration plus 1 cycle that ensures that 90% of samples are within this band.



Figure A6.3.2 An example of Symmetrical Components comparison and errors calculated with single cycle moving window.